

Chương 9: Số Học Cơ Bản Cho CP

[THUẬT NGỮ KHOA HỌC] Số Học Đồng Dư & Phân Rã Nhị Phân

Từ nguyên học: Các định lý số học kinh điển bắt nguồn từ Euclid (khoảng 300 TCN) và Carl Friedrich Gauss (1801, *Disquisitiones Arithmeticae*) đặt nền tảng cho mật mã học và khoa học máy tính hiện đại.

Mô hình trực giác: Khi giải bài toán số học lớn trong lập trình thi đấu, ta không thể lưu trữ các con số hàng triệu chữ số trong biến số thông thường. Số học đồng dư (Modulo) biến không gian số vô hạn thành chiếc đồng hồ hữu hạn tuần hoàn, và phân rã nhị phân chia nhỏ số mũ khổng lồ thành $O(\log N)$ bước nhân đơn giản.

9.1 Phép Toán Đồng Dư (Modulo) & Tránh Tràn Số

Trong lập trình thi đấu, các bài toán đếm hoặc tính toán lũy thừa thường yêu cầu chia lấy dư cho một số nguyên tố lớn, phổ biến nhất là $M = 10^9 + 7$.

Các tính chất đồng dư căn bản:

- $(A + B) \equiv ((A \bmod M) + (B \bmod M)) \bmod M$
- $(A - B) \equiv ((A \bmod M) - (B \bmod M) + M) \bmod M$ (Lưu ý cộng thêm M để tránh số âm trong C++)
- $(A \times B) \equiv ((A \bmod M) \times (B \bmod M)) \bmod M$

[DANGER] Cảnh báo tràn số khi nhân hai số 32-bit

Nếu $A, B < 10^9 + 7$, tích $A \times B$ có thể đạt tới xấp xỉ 10^{18} , vượt xa ngưỡng cực đại của kiểu `int` 32-bit ($\approx 2 \times 10^9$). Nếu viết `(a * b) % m`, CPU sẽ thực hiện phép nhân 32-bit dẫn đến tràn số trước khi lấy modulo. Bắt buộc phải ép kiểu 64-bit:

```
long long prod = (1LL * a * b) % m;
```

9.2 Thuật Toán Euclid: Ước Chung Lớn Nhất & Bội Chung Nhỏ Nhất

Thuật toán Euclid dựa trên bổ đề toán học:

$$\gcd(a, b) = \gcd(b, a \bmod b) \quad \text{với } b > 0, \quad \text{và } \gcd(a, 0) = a$$

Do $a \bmod b < \frac{a}{2}$ sau mỗi hai bước, thuật toán kết thúc sau tối đa $O(\log(\min(a, b)))$ bước chia dư.

Bội chung nhỏ nhất (LCM) được tính qua công thức:

$$\text{lcm}(a, b) = \frac{a \times b}{\gcd(a, b)} = \left(\frac{a}{\gcd(a, b)} \right) \times b$$

[TIP] Thực hiện phép chia trước để chống tràn số

Luôn tính $(a / \gcd(a, b)) * b$ thay vì $(a * b) / \gcd(a, b)$. Vì $\gcd(a, b)$ là ước của a , phép chia luôn là phép chia hết và thương luôn nhỏ hơn hoặc bằng a .

9.3 Lũy Thừa Nhị Phân (Binary Exponentiation)

Để tính $A^B \bmod M$ với $B \leq 10^{\{18\}}$, việc nhân lặp lại B lần mất $O(B)$ thời gian (chắc chắn bị Time Limit Exceeded). Thuật toán lũy thừa nhị phân phân tích B theo hệ cơ số 2:

$$A^B = A^{\sum b_i 2^i} = \prod_{b_i=1} A^{2^i}$$

Tại mỗi bước, cơ số được bình phương: $A \leftarrow A^2 \bmod M$, và nếu bit cuối của số mũ là 1, ta nhân cơ số hiện tại vào kết quả tích lũy.

[MẪU XẺ CÚ PHÁP C++14]

```
long long res = 1; while (b > 0) { if (b & 1) res = (1LL * res * a) % m; a = (1LL * a * a) % m; b >>= 1; }
```

Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>while (b > 0)</code>	Lặp chừng nào số mũ b vẫn còn bit 1 chưa duyệt.
<code>if (b & 1)</code>	Kiểm tra bit có trọng số nhỏ nhất của b có bật hay không.
<code>res = (1LL * res * a) % m;</code>	Nhân cơ số hiện tại vào tích lũy và lấy dư an toàn.
<code>a = (1LL * a * a) % m;</code>	Bình phương cơ số chuẩn bị cho lũy thừa 2 kế tiếp.
<code>b >>= 1;</code>	Dịch phải 1 bit (tương đương chia nguyên cho 2).

9.4 Sàng Số Nguyên Tố Eratosthenes

Để tìm mọi số nguyên tố nhỏ hơn hoặc bằng N , ta khởi tạo mảng đánh dấu `is_prime` toàn giá trị `true`. Duyệt p từ 2: nếu p là số nguyên tố, ta đánh dấu toàn bộ bội số của p bắt đầu từ $p \times p$ là hợp số (`false`).

Độ phức tạp tính toán là:

$$\sum_{p \leq N} \frac{N}{p} = O(N \log \log N)$$

9.5 Bảng Chạy Bàn Trên Giấy: Tính $3^5 \bmod 7$

Bước	Số mũ b	Bit chẵn/lẻ	Cơ số a	Biến tích lũy res
Khởi tạo	5 (101)	-	3	1
1	5	Lẻ (b & 1 = 1)	3	res = (1 * 3) % 7 = 3
nh phương 2 (b >> 1)		-	a = (3 * 3) % 7 = 2	3
	2	Chẵn (b & 1 = 0)	2	giữ nguyên res = 3
nh phương 1 (b >> 1)		-	a = (2 * 2) % 7 = 4	3
	1	Lẻ (b & 1 = 1)	4	res = (3 * 4) % 7 = 5
Kết thúc	0	-	a = (4 * 4) % 7 = 2	*5*

9.6 Bài Tập Tiêu Chuẩn

Bài toán: Bài 9.1 - Số Học Cơ Bản Cho CP (Number Theory Fundamentals)

(Nguồn: Nền tảng C++14 - Số học thuật toán)

Cho ba số nguyên dương A, B, M ($1 \leq A, B \leq 10^9, 2 \leq M \leq 10^9 + 7$) và số nguyên K ($1 \leq K \leq 100$). Em hãy thực hiện tính toán:

- Ước chung lớn nhất $G = \gcd(A, B)$ bằng thuật toán Euclid.
- Lũy thừa nhị phân $P = A^B \bmod M$.
- Đếm số lượng các số nguyên tố S nhỏ hơn hoặc bằng K bằng sàng Eratosthenes.

Đầu vào (Input): Bốn số nguyên A, B, M, K trên cùng một dòng.

Đầu ra (Output): In ra 3 số nguyên G, P, S cách nhau một khoảng trắng.

Ví dụ mẫu:

Đầu vào (Input):

3 5 7 10

Đầu ra (Output):

1 5 4

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): *Đáp án đối chiếu xem tại Phụ lục cuối sách*

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	3 5 7 10	[.....]

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
2	Warm-up	12 18 100 20	[.....]
3	Warm-up	4 6 10 5	[.....]
4	Edge case	1 1 13 1	[.....]
5	Edge case	7 7 7 2	[.....]
6	Edge case	15 1 1000 1	[.....]
7	Trick case	10 3 5 15	[.....]
8	Trick case	2 10 1000000007 25	[.....]
9	Mental challenge	6 8 11 8	[.....]
10	Mental challenge	5 4 13 7	[.....]

Chương 10: Cấu Trúc Dữ Liệu Tuần Tự STL

[THUẬT NGỮ KHOA HỌC] Chi Phí Khấu Hao (Amortized Time) & Vector Bộ Nhớ Động

Từ nguyên học: Được Robert Tarjan giới thiệu năm 1985 trong phân tích cấu trúc dữ liệu, kỹ thuật phân tích khấu hao đánh giá tổng chi phí của một chuỗi thao tác thay vì chỉ nhìn vào một thao tác đơn lẻ trong trường hợp xấu nhất.

Mô hình trực giác: Hãy tưởng tượng bạn mua một cuốn sổ ghi chép. Bạn viết từng dòng rất nhanh tốn 1 giây ($O(1)$). Khi hết trang, bạn phải đi mua cuốn sổ to gấp đôi và chép lại toàn bộ ($O(N)$). Dù thỉnh thoảng có một lần đi mua sổ tốn kém, nhưng trung bình mỗi dòng bạn viết vẫn chỉ tốn vài giây. Đó chính là bản chất Amortized $O(1)$ của `std::vector`.

10.1 Bản Chất Phần Cứng Của `std::vector`

`std::vector` là một mảng động tự co giãn được lưu trữ trong vùng nhớ Heap liên tục. Về mặt phần cứng và bộ nhớ, một đối tượng `vector` thường chỉ chiếm 24 bytes trên kiến trúc 64-bit, bao gồm đúng 3 con trỏ:

- `_M_start`: Trỏ tới phần tử đầu tiên của mảng.
- `_M_finish`: Trỏ tới vị trí liền kề sau phần tử cuối cùng hiện hữu. (Do đó, `size() = _M_finish - _M_start`).
- `_M_end_of_storage`: Trỏ tới giới hạn ô nhớ đã cấp phát của khối nhớ. (Do đó, `capacity() = _M_end_of_storage - _M_start`).

[TRỰC GIÁC HÌNH TƯỢNG] Phân biệt rạch ròi giữa `size()` và `capacity()`

- **Size (Kích thước thực tế):** Số lượng phần tử hiện đang được chương trình lưu giữ và sử dụng.
- **Capacity (Dung lượng đệm):** Tổng số ô nhớ liên tiếp đã được hệ điều hành cấp phát sẵn. Chỉ khi `size() == capacity()`, việc chèn thêm một phần tử mới kích hoạt việc cấp phát lại (reallocation).

10.2 Chiến Lược Nhân Đôi Dung Lượng & Chứng Minh Amortized $O(1)$

Khi ta gọi `push_back()` vào một vector đã đầy (`size == capacity`):

1. Vector xin hệ điều hành cấp phát một vùng nhớ mới có dung lượng gấp đôi ($2 \times \text{capacity}$).
2. Sao chép toàn bộ N phần tử cũ từ vùng nhớ cũ sang vùng nhớ mới.
3. Giải phóng vùng nhớ cũ.
4. Thêm phần tử mới vào cuối.

Chứng minh bằng Phương pháp Thế năng (Potential Method): Đặt thế năng $\Phi = 2 \times \text{size} - \text{capacity}$.

- Khi vector vừa được nhân đôi dung lượng ($\text{capacity} = 2N, \text{size} = N$): $\Phi = 2N - 2N = 0$.
- Trong mỗi thao tác `push_back` bình thường không cấp phát lại: chi phí thực tế là 1, `size` tăng 1, do đó Φ tăng thêm 2. Chi phí khấu hao $c_i' = c_i + \Delta\Phi = 1 + 2 = 3 = O(1)$.
- Khi vector đầy ($\text{size} = 2N, \text{capacity} = 2N$): $\Phi = 4N - 2N = 2N$.

- Thao tác thứ $2N + 1$ gây ra cấp phát lại: chi phí thực tế là $2N + 1$ (sao chép $2N$ phần tử và chèn 1 phần tử mới). Dung lượng mới là $4N$, kích thước mới là $2N + 1$. Thế năng mới $\Phi' = 2(2N + 1) - 4N = 2$. Độ biến thiên thế năng $\Delta\Phi = 2 - 2N$. Chi phí khấu hao: $c' = (2N + 1) + (2 - 2N) = 3 = O(1)$.

Như vậy, chi phí khấu hao cho mọi thao tác `push_back` luôn được chặn trên bởi hằng số $O(1)$.

10.3 Cặp Giá Trị `std::pair` & Bộ Ba `std::tuple`

Để lưu trữ các bộ dữ liệu đi liền nhau trong CP mà không cần khai báo `struct` dài dòng, ta sử dụng:

- `std::pair<T1, T2>`: Lưu cặp hai phần tử truy xuất qua `.first` và `.second`.
- `std::tuple<T1, T2, T3>`: Lưu bộ nhiều phần tử truy xuất qua `std::get<index>(t)`.

[MỖ XẼ CÚ PHÁP C++14]	
<pre>std::pair<int, int> p = {3, 5}; auto t = std::make_tuple(1, 2, 3); int x, y, z; std::tie(x, y, z) = t;</pre>	
Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>std::pair<int, int> p = {3, 5};</code>	Khởi tạo cặp giá trị với first=3, second=5.
<code>auto t = std::make_tuple(1, 2, 3);</code>	Tạo tuple gồm 3 giá trị nguyên.
<code>std::tie(x, y, z) = t;</code>	Giải nén (unpack) tuple vào các biến x, y, z bằng <code>std::tie</code> .

10.4 Bảng Chạy Bàn Trên Giấy: Theo Dõi Bộ Đệm Vector

Bước	Thao tác	Dữ liệu	Size	Capacity
0	Khởi tạo vector rỗng	<code>[]</code>	0	0
1	<code>push_back({1, 2})</code>	<code>[[{1,2}]]</code>	1	1 (cấp phát ban đầu)
2	<code>push_back({3, 4})</code>	<code>[[{1,2}, {3,4}]]</code>	2	2 (nhân đôi)
3	<code>pop_back()</code>	<code>[[{1,2}]]</code>	1	2 (capacity giữ nguyên)
4	<code>push_back({2, 1})</code>	<code>[[{1,2}, {2,1}]]</code>	2	2 (dùng lại slot trống)

10.5 Bài Tập Tiêu Chuẩn

Bài toán: Bài 10.1 - Điểm Tọa Độ Vector & Chi Phí Khấu Hao

(Nguồn: Nền tảng C++14 - Cấu trúc dữ liệu STL)

Một hệ thống quản lý danh sách các điểm 2D lưu trong `std::vector<std::pair<int, int>>`. Ban đầu vector rỗng. Cho số nguyên N ($1 \leq N \leq 20$). Dòng tiếp theo gồm N điểm (x_i, y_i) . Với mỗi điểm:

- Nếu tổng tọa độ $x_i + y_i \geq 0$, thực hiện nạp điểm vào cuối vector (`push_back`).
- Nếu tổng tọa độ $x_i + y_i < 0$, thực hiện loại bỏ điểm gần nhất vừa thêm vào (`pop_back`) nếu vector không rỗng.

Sau khi xử lý hết N điểm, in ra trên một dòng: số phần tử còn lại trong vector `size`, hoành độ nhỏ nhất `min_x` và tung độ lớn nhất `max_y` trong các điểm còn lại. (Nếu vector rỗng in `0 0 0`).

Đầu vào (Input): Dòng 1 ghi N . N dòng tiếp theo mỗi dòng ghi hai số nguyên x_i, y_i .

Đầu ra (Output): In ra 3 số nguyên `size min_x max_y` trên một dòng.

Ví dụ mẫu:

Đầu vào (Input):

4
1 2
3 4
-5 -2
2 1

Đầu ra (Output):

2 1 2

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): <i>Đáp án đối chiếu xem tại Phụ lục cuối sách</i>			
Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	4 1 2 3 4 -5 -2 2 1	[.....]
2	Warm-up	3 5 5 2 8 -1 4	[.....]
3	Warm-up	2 0 0 4 -2	[.....]
4	Edge case	1 -3 5	[.....]
5	Edge case	1 -2 -3	[.....]
6	Edge case	4 1 1 2 2 -3 -1 -4 -4	[.....]
7	Trick case	4 -1 -1 -2 -2 3 3 4 4	[.....]

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
8	Trick case	<pre> 5 1 0 -1 -2 2 0 -3 -3 5 5 </pre>	[.....]
9	Mental challenge	<pre> 6 2 3 1 5 -4 -1 3 1 0 2 -5 -2 </pre>	[.....]
10	Mental challenge	<pre> 5 -2 2 4 -1 -1 7 -6 2 3 0 </pre>	[.....]

Chương 11: Kỹ Thuật Hai Con Trỏ & Cửa Sổ Trượt

[THUẬT NGỮ KHOA HỌC] Hai Con Trỏ & Tính Đơn Điều (Monotonicity)

Từ nguyên học: Xuất hiện như một phương pháp tư duy tối ưu cơ bản trong các thuật toán sắp xếp và tìm kiếm từ thập niên 1970, kỹ thuật Hai Con Trỏ khai thác triệt để tính chất đơn điệu của dãy số hoặc hàm mục tiêu.

Mô hình trực giác: Thay vì duyệt hai vòng lặp lồng nhau $O(N^2)$ thử mọi cặp đoạn con, ta duy trì hai chỉ số L và R . Khi biên phải R dịch chuyển về phía trước làm tăng chi phí, biên trái L cũng chỉ dịch chuyển một chiều theo sau để thiết lập lại điều kiện hợp lệ. Nhờ đó mỗi phần tử chỉ được thăm tối đa hai lần, biến thuật toán từ $O(N^2)$ thành $O(N)$.

11.1 Hai Mô Hình Hai Con Trỏ Cốt Lõi

- Hai Con Trỏ Chạy Ngược Chiều (Opposite Direction / Converging Pointers):** Áp dụng trên mảng đã được sắp xếp tăng dần. Một con trỏ xuất phát từ đầu mảng ($L = 0$), một con trỏ xuất phát từ cuối mảng ($R = N - 1$).
 - Nếu tổng $A[L] + A[R] < S$: muốn tăng tổng, bắt buộc phải tăng L ($L++$).
 - Nếu tổng $A[L] + A[R] > S$: muốn giảm tổng, bắt buộc phải giảm R ($R--$).
 - Nếu $A[L] + A[R] == S$: ghi nhận nghiệm.
- Cửa Sổ Trượt Cùng Chiều (Sliding Window / Same Direction):** Áp dụng khi tìm đoạn con liên tiếp $[L, R]$ thỏa mãn một ràng buộc đơn điệu (chẳng hạn tổng các phần tử dương $\leq S$, hoặc không quá K phần tử phân biệt). Biên phải R mở rộng cửa sổ từng bước. Khi tổng vượt ngưỡng, biên trái L co hẹp lại.

[LÝ THUYẾT] Bất biến vòng lặp (Loop Invariant) của Cửa Sổ Trượt

Tại cuối mỗi bước lặp của R :

- Đoạn $[L, R]$ luôn là đoạn con dài nhất kết thúc tại vị trí R có tổng không vượt quá S .
- Mọi đoạn con bắt đầu từ $i \in [L, R]$ và kết thúc tại R đều có tổng $\leq S$. Do đó, số lượng đoạn con hợp lệ kết thúc tại R chính xác bằng $(R - L + 1)$.

11.2 Mô Xê Cú Pháp Cửa Sổ Trượt C++14

[MÔ XÊ CÚ PHÁP C++14]	
<pre>for (int R = 0; R < n; ++R) { sum += a[R]; while (sum > s && L ≤ R) { sum -= a[L++]; } total += (R - L + 1); }</pre>	
Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>for (int R = 0; R < n; ++R)</code>	Duyệt biên phải R tuần tự từ đầu đến cuối mảng.
<code>sum += a[R];</code>	Hấp thụ phần tử mới vào cửa sổ hiện tại.
<code>while (sum > s && L ≤ R)</code>	Kiểm tra điều kiện vi phạm ngưỡng tổng S.
<code>sum -= a[L++];</code>	Thu hẹp biên trái L và cập nhật lại biến tổng tích lũy.
<code>total += (R - L + 1);</code>	Cộng số lượng đoạn con hợp lệ kết thúc tại R vào tổng số nghiệm.

11.3 Bảng Chạy Bàn Trên Giấy: Mảng [2, 1, 3, 4, 1] Với Ngưỡng $S = 7$

Bước R	Phần tử thêm	Thao tác L	Cửa sổ [L, R]	Tổng sum	Đoạn con kết thúc tại R
R=0	a[0]=2	L=0 giữ nguyên	[0, 0] = [2]	2 ≤ 7	1: [2]
R=1	a[1]=1	L=0 giữ nguyên	[0, 1] = [2, 1]	3 ≤ 7	2: [1], [2, 1]
R=2	a[2]=3	L=0 giữ nguyên	[0, 2] = [2, 1, 3]	6 ≤ 7	3: [3], [1, 3], [2, 1, 3]
R=3	a[3]=4	L tăng: 0 -> 1 -> 2	[2, 3] = [3, 4]	7 ≤ 7	2: [4], [3, 4]
R=4	a[4]=1	L tăng: 2 -> 3	[3, 4] = [4, 1]	5 ≤ 7	2: [1], [4, 1]
Tổng kết	-	-	Max len = 3	-	*Tổng = 1 + 2 + 3 + 2 + 2 = 10*

11.4 Bài Tập Tiêu Chuẩn

Bài toán: Bài 11.1 - Cửa Sổ Trượt: Đoạn Con Liên Tiếp Có Tổng Không Vượt Quá S

(Nguồn: Nền tảng C++14 - Kỹ thuật Hai Con Trỏ)

Cho dãy N số nguyên dương $A_1, A_2, \dots, A_N$ và một số nguyên dương S . Sử dụng kỹ thuật Cửa Sổ Trượt (Hai Con Trỏ L, R cùng chiều), hãy tìm:

- Độ dài lớn nhất của một đoạn con liên tiếp có tổng các phần tử $\leq S$.
- Tổng số lượng đoạn con liên tiếp có tổng $\leq S$.

Nếu không có đoạn con nào thỏa mãn (mọi phần tử $> S$), in ra `0 0`.

Đầu vào (Input): Dòng 1 ghi hai số nguyên N, S ($1 \leq N \leq 20, 1 \leq S \leq 10^9$). Dòng 2 ghi N số nguyên $A_1, A_2, \dots, A_N$ ($A_i \geq 1$).

Đầu ra (Output): In ra hai số nguyên `max_len total_count` trên một dòng.

Ví dụ mẫu:

Đầu vào (Input):

5 7

2 1 3 4 1

Đầu ra (Output):

3 10

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): *Đáp án đối chiếu xem tại Phụ lục cuối sách*

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	5 7 2 1 3 4 1	
2	Warm-up	4 5 1 2 1 1	
3	Warm-up	4 3 1 1 1 1	
4	Edge case	1 10 5	
5	Edge case	1 5 10	
6	Edge case	3 2 5 7 9	
7	Trick case	5 4 2 10 3 1 10	
8	Trick case	3 6 1 2 3	
9	Mental challenge	6 6 3 2 1 4 2 1	
10	Mental challenge	6 5 2 2 2 2 2 2	

Chương 12: Sắp Xếp & Tìm Kiếm Nhị Phân, Chặt Nhị Phân Kết Quả

[THUẬT NGỮ KHOA HỌC] Không Gian Nghiệm Đơn Điều & Chặt Nhị Phân Kết Quả

Từ nguyên học: *Tìm kiếm nhị phân xuất hiện từ thời Babylon cổ đại trong thuật toán xấp xỉ nghiệm và được chuẩn hóa trong khoa học máy tính bởi John Mauchly (1946).*

Mô hình trực giác: Khi gặp bài toán 'tìm giá trị nhỏ nhất thỏa mãn điều kiện X' mà việc giải trực tiếp quá khó, nhưng nếu cho trước một giá trị nghiệm mid cụ thể thì ta lại kiểm tra được dễ dàng trong $O(N)$. Nếu hàm kiểm tra có tính chất đơn điệu (False False ... True True), ta có thể chặt đôi không gian nghiệm qua mỗi bước $\log_2(\text{Range})$, biến bài toán tối ưu thành bài toán kiểm tra tính khả thi.

12.1 `std::sort` Trong C++14 & Quy Tắc Strict Weak Ordering

Thuật toán `std::sort` trong Thư viện Tiêu chuẩn C++ được cài đặt bằng kỹ thuật Introsort (kết hợp QuickSort, HeapSort và InsertionSort), đảm bảo thời gian chạy $O(N \log N)$ trong mọi trường hợp xấu nhất.

[DANGER] Cạm bẫy sống còn: Strict Weak Ordering

Hàm so sánh (comparator) truyền vào `std::sort` bắt buộc phải tuân thủ tiên đề **Strict Weak Ordering**. Quy tắc quan trọng nhất: Khi hai phần tử bằng nhau ($a == b$), hàm so sánh **bắt buộc phải trả về false!** Nếu viết `return a ≤ b;`, `std::sort` có thể rơi vào vòng lặp vô tận hoặc truy xuất vượt biên bộ nhớ (Segmentation Fault).

12.2 C++14 Generic Lambdas Làm Hàm So Sánh

C++14 cho phép sử dụng `auto` trong danh sách tham số của hàm ẩn danh (lambda), giúp cú pháp so sánh ngắn gọn và uyển chuyển:

```
std::sort(pts.begin(), pts.end(), [](const auto& a, const auto& b) {
    if (a.first != b.first) return a.first < b.first;
    return a.second > b.second; // Tung độ giảm dần nếu hoành độ bằng nhau
});
```

12.3 Tìm Kiếm Nhị Phân: `lower_bound` vs `upper_bound`

Trên một mảng hoặc vector đã được sắp xếp tăng dần:

- `std::lower_bound(first, last, val)`: Trả về con trỏ/iterator tới phần tử đầu tiên có giá trị **lớn hơn hoặc bằng** x ($\geq x$).
- `std::upper_bound(first, last, val)`: Trả về con trỏ/iterator tới phần tử đầu tiên có giá trị **lớn hơn hẳn** x ($> x$).

Số lượng phần tử có giá trị bằng x chính là khoảng cách `upper_bound - lower_bound`.

12.4 Chặt Nhị Phân Kết Quả (Binary Search on Answer)

Giả sử cần tìm giá trị nhỏ nhất X sao cho hàm khả thi `check(X) == true`.

- Xác định cận dưới chắc chắn không thỏa mãn hoặc cận cực tiểu: $L = \max(A)$.
- Xác định cận trên chắc chắn thỏa mãn: $R = \sum A$.
- Tính điểm giữa: $M = L + \frac{R-L}{2}$. (Cách viết chống tràn số).

[MÔ XẼ CÚ PHÁP C++14]

```
long long low = max_val, high = sum_val, ans = high; while (low ≤ high) { long
long mid = low + (high - low) / 2; if (check(mid)) { ans = mid; high = mid - 1; }
else { low = mid + 1; } }
```

Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>while (low ≤ high)</code>	Vòng lặp thu hẹp không gian tìm kiếm chừng nào khoảng còn hợp lệ.
<code>mid = low + (high - low) / 2;</code>	Tính giá trị thử nghiệm chính giữa tránh tràn số nguyên.
<code>if (check(mid))</code>	Kiểm tra nghiệm mid có thỏa mãn chia công việc thành $\leq K$ đoạn không.
<code>ans = mid; high = mid - 1;</code>	Nếu khả thi, ghi nhận đáp án tốt hơn và tìm tiếp ở nửa bên trái.
<code>else { low = mid + 1; }</code>	Nếu không khả thi, bắt buộc phải tăng tải trọng ở nửa bên phải.

12.5 Bảng Chạy Bàn Trên Giấy: Mảng $[2, 4, 7, 1, 6]$ Với $K = 3$

Khoảng tìm kiếm ban đầu: $L = \max(A) = 7$, $R = \sum A = 20$.

Lần lặp	low	high	mid	check(mid)	Hành động cập nhật
1	7	20	13	True (2 đoạn ≤ 3)	ans = 13, high = 13 - 1 = 12
2	7	12	9	True (3 đoạn ≤ 3)	ans = 9, high = 9 - 1 = 8

Lần lặp	low	high	mid	check(mid)	Hành động cập nhật
3	7	8	7	True (3 đoạn ≤ 3)	ans = 7, high = 7 - 1 = 6
Kết thúc	7	6	-	-	*Đáp án tối ưu = 7, Số lần check = 3*

12.6 Bài Tập Tiêu Chuẩn

Bài toán: Bài 12.1 - Chặt Nhị Phân Phân Đoạn Công Việc (Workload Partitioning)

(Nguồn: Nền tảng C++14 - Chặt nhị phân kết quả)

Cho dãy N công việc với thời gian hoàn thành $A_1, A_2, \dots, A_N$ và K công nhân. Cần phân chia dãy công việc liên tiếp này cho K công nhân sao cho thời gian làm việc của công nhân làm nhiều nhất là nhỏ nhất có thể. Mỗi công nhân nhận một đoạn con liên tiếp các công việc.

Em hãy tìm:

- Giá trị tải trọng tối thiểu khả thi `ans`.
- Số lần hàm `check(mid)` được gọi trong quá trình chặt nhị phân `check_count`.

Đầu vào (Input): Dòng 1 ghi hai số nguyên N, K ($1 \leq N \leq 20, 1 \leq K \leq N$). Dòng 2 ghi N số nguyên dương $A_1, A_2, \dots, A_N$.

Đầu ra (Output): In ra hai số nguyên `ans check_count` cách nhau một khoảng trắng.

Ví dụ mẫu:

Đầu vào (Input):

```
5 3
2 4 7 1 6
```

Đầu ra (Output):

```
7 3
```

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): Đáp án đối chiếu xem tại Phụ lục cuối sách

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	5 3 2 4 7 1 6	[.....]
2	Warm-up	4 2 10 20 30 40	[.....]
3	Warm-up	3 3 5 10 15	[.....]
4	Edge case	1 1 42	[.....]
5	Edge case	5 1 1 2 3 4 5	[.....]

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
6	Edge case	4 4 7 7 7 7	[.....]
7	Trick case	5 2 100 1 1 1 1	[.....]
8	Trick case	6 3 1 1 1 1 1 1	[.....]
9	Mental challenge	5 2 3 2 4 1 5	[.....]
10	Mental challenge	6 2 1 2 3 4 5 6	[.....]

Chương 13: Đệ Quy Sơ Cấp & Cây Gọi Đệ Quy

[THUẬT NGỮ KHOA HỌC] Đệ Quy & Cây Gọi Đệ Quy (Recursion Tree)

Từ nguyên học: Khái niệm hàm đệ quy được phát triển trong logic toán học bởi Thoralf Skolem (1923) và Kurt Gödel (1931), trước khi trở thành hòn đá tảng của khoa học máy tính thông qua phép chia để trị và quy hoạch động.

Mô hình trực giác: Một bài toán đệ quy giống như việc giải một câu đố bằng cách phân rã nó thành các câu đố nhỏ hơn có cùng cấu trúc, cho đến khi chạm tới câu đố nhỏ nhất mà câu trả lời là hiển nhiên (Base Case). Khi vẽ trên giấy, toàn bộ tiến trình này nở rộng thành một Cây gọi đệ quy phân nhánh.

13.1 Hai Trụ Cột Sinh Tử Của Hàm Đệ Quy

Mọi hàm đệ quy đúng đắn bắt buộc phải có hai thành phần:

- Điểm dừng cơ sở (Base Case):** Điều kiện quy mô bài toán đủ nhỏ để trả về kết quả ngay lập tức mà không gọi thêm hàm đệ quy nào khác.
- Bước đệ quy (Recursive Step):** Lời gọi hàm chính nó với tham số tiến dần về điểm dừng cơ sở.

[DANGER] Thảm họa tràn ngăn xếp (Stack Overflow)

Nếu một hàm đệ quy không có điểm dừng, hoặc tham số truyền vào không tiến về điểm dừng cơ sở, CPU sẽ liên tục nạp các khung ngăn xếp (stack frames) mới vào vùng nhớ Stack của hệ điều hành cho tới khi cạn kiệt bộ nhớ ($\approx 8\text{MB}$ trên Linux). Chương trình sẽ bị hủy diệt ngay lập tức bằng lỗi Crash / RTE.

13.2 Mô Hình Cây Gọi Đệ Quy Trên Giấy

Khi chạy bàn một hàm đệ quy trên giấy không có máy tính:

- Mỗi nút tròn đại diện cho một lần gọi hàm với tham số cụ thể, ví dụ $f(3)$.
- Nhánh từ nút cha xuống nút con biểu thị lời gọi hàm con được kích hoạt từ thân hàm cha.
- Thứ tự duyệt trên cây: CPU thực thi theo chiều sâu (Depth-First Search): đi hết nhánh con bên trái (Pre-order), thu nhận giá trị trả về (Post-order), rồi mới chuyển sang nhánh con bên phải.

13.3 Mổ Xẻ Cú Pháp Hàm Đệ Quy Phân Hoạch

Xét hàm $f(n) = f(n-1) + f(\frac{n}{2}) + n$ với $f(n) = 1$ khi $n \leq 1$:

[MỔ XẺ CÚ PHÁP C++14]	
<pre>long long solve_f(int n) { if (n ≤ 1) return 1; return solve_f(n - 1) + solve_f(n / 2) + n; }</pre>	
Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>if (n ≤ 1) return 1;</code>	Điểm dừng cơ sở: khi n=0 hoặc n=1, trả về 1 ngay lập tức.
<code>solve_f(n - 1)</code>	Nhánh trái: đệ quy giải bài toán quy mô n - 1.
<code>solve_f(n / 2)</code>	Nhánh phải: đệ quy giải bài toán quy mô thu nhỏ phân nửa n / 2.
<code>+ n;</code>	Tổng hợp kết quả bài toán con cộng với chi phí gộp của bài toán hiện tại.

13.4 Bảng Chạy Bàn Trên Giấy: Cây Đệ Quy Cho $N = 3$

Lời gọi gốc $f(3)$:

- $f(3)$ gọi $f(2)$ và $f(1)$
 - $f(2)$ gọi $f(1)$ và $f(1)$
 - $f(1)$ là base case, trả về 1 (lời gọi 1)
 - $f(1)$ là base case, trả về 1 (lời gọi 2)
 - $f(2) = 1 + 1 + 2 = 4$
 - $f(1)$ là base case, trả về 1 (lời gọi 3)
- $f(3) = 4 + 1 + 3 = 8$

Thứ tự gọi	Hàm f(n)	Độ sâu	Hành động	Giá trị trả về
1	f(3)	0	Gọi con f(2) và f(1)	Chờ nhánh con
2	f(2)	1	Gọi con f(1) và f(1)	Chờ nhánh con
3	f(1)	2	Base case $n \leq 1$	1
4	f(1)	2	Base case $n \leq 1$	1
Thu nhận f(2)	f(2)	1	$1 + 1 + 2$	4
5	f(1)	1	Base case $n \leq 1$	1
Thu nhận f(3)	f(3)	0	$4 + 1 + 3$	*8*

13.5 Bài Tập Tiêu Chuẩn

Bài toán: Bài 13.1 - Cây Gọi Đệ Quy: Phân Hoạch & Đếm Nút (Recursion Tree Node Counting) (Nguồn: Nền tảng C++14 - Đệ quy cơ bản)

Xét hàm đệ quy $f(n)$ được định nghĩa như sau:

- Nếu $n \leq 1$: $f(n) = 1$.
- Nếu $n > 1$: $f(n) = f(n-1) + f(\frac{n}{2}) + n$ (với phép chia nguyên $(/)$).

Cho số nguyên N ($0 \leq N \leq 15$). Em hãy tính nhẩm trên giấy:

- Giá trị $f(N)$.
- Tổng số lời gọi hàm đệ quy (tổng số nút trong cây đệ quy) được tạo ra `total_calls`.
- Chiều sâu tối đa của cây đệ quy `max_depth` (gốc có độ sâu 0).

Đầu vào (Input): Một số nguyên duy nhất N trên một dòng.

Đầu ra (Output): In ra 3 số nguyên `val total_calls max_depth` cách nhau một khoảng trắng.

Ví dụ mẫu:

Đầu vào (Input):

3

Đầu ra (Output):

8 5 2

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): Đáp án đối chiếu xem tại Phụ lục cuối sách

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	2	[.....]
2	Warm-up	3	[.....]
3	Warm-up	4	[.....]
4	Edge case	0	[.....]
5	Edge case	1	[.....]
6	Edge case	5	[.....]
7	Trick case	6	[.....]
8	Trick case	7	[.....]
9	Mental challenge	8	[.....]
10	Mental challenge	3	[.....]

Chương 14: Thuật Toán Quay Lui & Nhánh Cận

[THUẬT NGỮ KHOA HỌC] Quay Lui & Nhánh Cận (Backtracking & Branch-and-Bound)

Từ nguyên học: D. H. Lehmer đưa ra thuật ngữ *Backtracking* vào những năm 1950, và A. H. Land cùng A. G. Doig (1960) đề xuất phương pháp *Branch-and-Bound* trong tối ưu hóa tổ hợp rời rạc.

Mô hình trực giác: Hãy tưởng tượng bạn đang thám hiểm một mê cung nhiều ngã rẽ. Bạn đi theo một con đường; nếu gặp ngõ cụt hoặc biển báo 'đường cụt chắc chắn' (tia nhánh), bạn quay lui lại ngã ba gần nhất và thử lối đi tiếp theo. Thuật toán quay lui duyệt toàn bộ cây không gian trạng thái theo chiều sâu nhưng biết dừng lại kịp thời để không lãng phí thời gian vào những nhánh vô vọng.

14.1 Bản Chất Của Bộ Ba Thao Tác Quay Lui

Khung sườn của mọi bài toán quay lui luôn tuân thủ chu trình 4 pha:

- Kiểm tra hoàn tất:** Nếu đã chọn đủ N thành phần nghiệm, ghi nhận lời giải.
- Thử các khả năng:** Duyệt qua mọi ứng viên hợp lệ cho bước hiện tại.
- Đánh dấu & Gọi đệ quy:** Ghi nhận ứng viên đã dùng (nạp trạng thái), rồi gọi đệ quy bước $i + 1$.
- Hoàn tác (Backtrack):** Khi lời gọi con kết thúc, **hủy bỏ đánh dấu** (trả trạng thái về như cũ) để các nhánh sau sử dụng lại.

14.2 Kỹ Thuật Tia Nhánh $O(1)$ Trong Bài Toán Xếp Hậu (N-Queens)

Xét bàn cờ $N \times N$. Đặt lần lượt từng quân hậu vào các hàng $0, 1, \dots, N - 1$. Để đảm bảo hai quân hậu không ăn nhau trong $O(1)$:

- Cùng cột:** Mảng đánh dấu `used_col[c]`.
- Cùng đường chéo chính (hướng $\searrow$):** Các ô có cùng giá trị $(r - c)$. Để tránh chỉ số âm, ta đánh số $(r - c + N)$: mảng `diag2[r - c + n]`.
- Cùng đường chéo phụ (hướng $\swarrow$):** Các ô có cùng tổng $(r + c)$: mảng `diag1[r + c]`.

[MÔ XẼ CÚ PHÁP C++14]

```
for (int c = 0; c < n; ++c) { int d1 = r + c, d2 = r - c + n; if (!used[c] && !diag1[d1] && !diag2[d2]) { used[c] = diag1[d1] = diag2[d2] = true; backtrack(r + 1); used[c] = diag1[d1] = diag2[d2] = false; } }
```

Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>int d1 = r + c, d2 = r - c + n;</code>	Tính chỉ số đường chéo chính và đường chéo phụ độc nhất.
<code>if (!used[c] && !diag1[d1] && !diag2[d2])</code>	Tia nhánh cận $O(1)$: chỉ duyệt nếu ô không bị chiếu tướng.
<code>used[c] = diag1[d1] = diag2[d2] = true;</code>	Nạp trạng thái: phong tỏa cột và 2 đường chéo.
<code>backtrack(r + 1);</code>	Đệ quy đặt quân hậu cho hàng tiếp theo $r + 1$.
<code>used[c] = diag1[d1] = diag2[d2] = false;</code>	HOÀN TÁC: dỡ bỏ phong tỏa cho các nhánh thử nghiệm kế tiếp.

14.3 Bảng Chạy Bàn Trên Giấy: Tìm Nghiệm Đầu Tiên Cho Bàn Cờ 4×4

Hàng r	Cột c thử	Kiểm tra chiếu tướng	Hành động	Trạng thái cột đã đặt
Hàng 0	<code>c=0 (ô 0,0)</code>	Hợp lệ	Đặt hậu tại $c=0$, xuống hàng 1	<code>[0, -, -, -]</code>
Hàng 1	<code>c=0, 1</code>	Bị chiếu	Bỏ qua	-
Hàng 1	<code>c=2 (ô 1,2)</code>	Hợp lệ	Đặt hậu tại $c=2$, xuống hàng 2	<code>[0, 2, -, -]</code>
Hàng 2	<code>c=0, 1, 2, 3</code>	Mọi ô đều bị chiếu!	Ngõ cụt! Quay lui hàng 1	-
Hàng 0	<code>c=1 (ô 0,1)</code>	Hợp lệ	Đặt hậu tại $c=1$, xuống hàng 1	<code>[1, -, -, -]</code>
Hàng 1	<code>c=3 (ô 1,3)</code>	Hợp lệ	Đặt hậu tại $c=3$, xuống hàng 2	<code>[1, 3, -, -]</code>
Hàng 2	<code>c=0 (ô 2,0)</code>	Hợp lệ	Đặt hậu tại $c=0$, xuống hàng 3	<code>[1, 3, 0, -]</code>

Hàng r	Cột c thử	Kiểm tra chiếu tướng	Hành động	Trạng thái cột đã đặt
Hàng 3	c=2 (ô 3,2)	Hợp lệ	Đặt hậu tại c=2 → *Hoàn tất!*	*Nghiệm: 2 4 1 3*

14.4 Bài Tập Tiêu Chuẩn

Bài toán: Bài 14.1 - Quay Lui Nhánh Cận: Bài Toán Xếp Hậu
Trên Bàn Cờ N x N

(Nguồn: Nền tảng C++14 - Thuật toán quay lui)

Cho số nguyên N ($1 \leq N \leq 8$). Hãy tìm số cách đặt N quân hậu trên bàn cờ vua $N \times N$ sao cho không có hai quân hậu nào ăn nhau. Đồng thời, in ra cấu hình nghiệm đầu tiên tìm được (theo thứ tự duyệt cột tăng dần từ hàng 1 đến hàng N , chỉ số cột từ 1 đến N). Nếu không có cách xếp nào thỏa mãn (như trường hợp $N = 2, 3$), in ra `0 NONE`.

Đầu vào (Input): Một số nguyên duy nhất N trên một dòng.

Đầu ra (Output): In ra số lượng nghiệm `solution_count` kèm theo cấu hình nghiệm đầu tiên trên cùng một dòng.

Ví dụ mẫu:

Đầu vào (Input):

`4`

Đầu ra (Output):

`2 2 4 1 3`

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì):

Đáp án đối chiếu xem tại Phụ lục cuối sách

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	<code>4</code>	[.....]
2	Warm-up	<code>1</code>	[.....]
3	Warm-up	<code>5</code>	[.....]
4	Edge case	<code>2</code>	[.....]
5	Edge case	<code>3</code>	[.....]
6	Edge case	<code>6</code>	[.....]
7	Trick case	<code>7</code>	[.....]
8	Trick case	<code>8</code>	[.....]
9	Mental challenge	<code>4</code>	[.....]
10	Mental challenge	<code>5</code>	[.....]

Chương 15: Thuật Toán Tham Lam & Phương Pháp Chứng Minh Trên Giấy

[THUẬT NGỮ KHOA HỌC] Thuật Toán Tham Lam & Luận Điểm Đổi Chỗ (Exchange Argument)

Từ nguyên học: Xuất hiện trong các nghiên cứu kinh điển về cấu trúc Matroid của Hassler Whitney (1935) và bài toán tìm cây khung nhỏ nhất của Kruskal (1956) và Prim (1957).

Mô hình trực giác: Một thuật toán tham lam luôn đưa ra lựa chọn có vẻ tốt nhất ở thời điểm hiện tại (cực trị cục bộ) với hy vọng dẫn tới nghiệm tối ưu toàn cục. Tuy nhiên, sự khác biệt giữa một thuật toán tham lam thiên tài và một thuật toán sai lầm nằm ở chứng minh toán học: ta phải chứng minh được rằng việc chọn phần tử tham lam không bao giờ ngăn cản ta đạt được kết quả tối ưu.

15.1 Khi Nào Tham Lam Đúng & Khi Nào Tham Lam Thất Bại?

Một bài toán có thể giải bằng thuật toán Tham lam nếu thỏa mãn hai tính chất:

- Tính chất lựa chọn tham lam (Greedy-choice Property):** Nghiệm tối ưu toàn cục có thể được xây dựng bằng cách đưa ra các lựa chọn tối ưu cục bộ tuần tự mà không cần quay lui hay xét lại quá khứ.
- Cấu trúc con tối ưu (Optimal Substructure):** Sau khi thực hiện lựa chọn tham lam, bài toán còn lại vẫn là bài toán con cùng dạng và nghiệm tối ưu của bài toán con kết hợp với lựa chọn tham lam tạo thành nghiệm tối ưu bài toán ban đầu.

[DANGER] Cảnh báo: Tham lam mù quáng

Bài toán Đổi tiền xu: Nếu hệ mệnh giá là $\{1, 5, 10, 20, 50\}$, thuật toán tham lam luôn đúng. Nhưng nếu hệ mệnh giá là $\{1, 3, 4\}$ và cần đổi số tiền là 6:

- Tham lam: chọn 4, còn 2 $\rightarrow$ chọn hai đồng 1 $\rightarrow$ Tổng cộng 3 đồng xu ($4 + 1 + 1$).
- Tối ưu thực tế: chọn hai đồng 3 $\rightarrow$ Chỉ mất 2 đồng xu ($3 + 3$).

Do đó, khi không có chứng minh toán học, tham lam rất dễ trở thành sai lầm chết người trong phòng thi!

15.2 Phương Pháp Chứng Minh Bằng Luận Điểm Đổi Chỗ (Exchange Argument)

Để chứng minh chiến lược tham lam $S = (s_1, s_2, \dots, s_k)$ là tối ưu trên giấy:

- Giả sử tồn tại một nghiệm tối ưu $OPT = (o_1, o_2, \dots, o_m)$ khác với S .
- Tìm phần tử đầu tiên mà OPT khác với S (giả sử tại bước i , $s_i \neq o_i$).
- Thay thế o_i bằng s_i trong OPT để tạo ra một nghiệm mới OPT' .
- Chứng minh rằng OPT' vẫn là một nghiệm hợp lệ và chất lượng của OPT' không hề kém hơn OPT ($|OPT'| \geq |OPT|$).
- Lặp lại lập luận bằng quy nạp toán học cho đến khi biến đổi hoàn toàn OPT thành S . Suy ra S cũng là một nghiệm tối ưu!

15.3 Bài Toán Lựa Chọn Hoạt Động (Interval Scheduling)

Cho N cuộc họp với thời gian bắt đầu S_i và kết thúc E_i . Ta cần chọn nhiều cuộc họp nhất không giao nhau. **Chiến lược tối ưu:** Luôn chọn cuộc họp có **thời điểm kết thúc sớm nhất** (E_i nhỏ nhất) tương thích với các cuộc họp trước đó.

[MỖ XẼ CÚ PHÁP C++14]	
<pre>std::sort(a.begin(), a.end(), [](const auto& u, const auto& v) { return u.e < v.e; }); for (const auto& item : a) { if (item.s ≥ last_end) { count++; last_end = item.e; } }</pre>	
Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>return u.e < v.e;</code>	Sắp xếp danh sách cuộc họp tăng dần theo thời điểm kết thúc.
<code>for (const auto& item : a)</code>	Duyệt tuần tự theo thứ tự tham lam.
<code>if (item.s ≥ last_end)</code>	Kiểm tra cuộc họp có xung đột thời gian với cuộc họp trước không.
<code>last_end = item.e;</code>	Cập nhật thời điểm kết thúc phòng họp bận rộn.

15.4 Bảng Chạy Bàn Trên Giấy: 4 Cuộc Họp [(1, 3), (2, 5), (3, 9), (6, 8)]

Sắp xếp theo thời gian kết thúc:

1. (1, 3) kết thúc lúc 3
2. (2, 5) kết thúc lúc 5
3. (6, 8) kết thúc lúc 8
4. (3, 9) kết thúc lúc 9

Cuộc họp	Khung giờ [S, E]	last_end hiện tại	Xung đột?	Quyết định
1	[1, 3]	-1	Không ($1 \geq -1$)	CHỌN (last_end = 3, dur += 2)
2	[2, 5]	3	Có ($2 < 3$)	LOẠI (trùng giờ)
3	[6, 8]	3	Không ($6 \geq 3$)	CHỌN (last_end = 8, dur += 2)
4	[3, 9]	8	Có ($3 < 8$)	LOẠI (trùng giờ)
Kết quả	-	-	-	*Chọn 2 cuộc họp, Tổng thời gian = 4*

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
6	Edge case	3 2 6 2 6 2 6	[.....]
7	Trick case	5 1 10 2 9 3 8 4 7 5 6	[.....]
8	Trick case	3 0 5 5 10 10 15	[.....]
9	Mental challenge	8 1 4 3 5 0 6 5 7 3 8 5 9 6 10 8 11	[.....]
10	Mental challenge	5 2 5 1 3 4 7 6 9 8 10	[.....]

Chương 16: Ngăn Xếp & Hàng Đội STL

[THUẬT NGỮ KHOA HỌC] Cấu Trúc LIFO, FIFO & Ngăn Xếp Đơn Điều (Monotonic Stack)

Từ nguyên học: Alan Turing giới thiệu khái niệm ngăn xếp (Stack) năm 1946 với lệnh 'bury' và 'unbury' để lưu vết địa chỉ quay về của chương trình con.

Mô hình trực giác: Ngăn xếp (Stack) giống như một chồng đĩa: chiếc đĩa đặt vào sau cùng sẽ được lấy ra đầu tiên (LIFO - Last In First Out). Ngăn xếp Đơn điệu là kỹ thuật duy trì các phần tử trong ngăn xếp luôn tăng dần hoặc giảm dần bằng cách loại bỏ các phần tử bị vi phạm tính đơn điệu, giải quyết các bài toán tìm cực trị kế cận trong thời gian tuyến tính $O(N)$.

16.1 Toàn Cảnh Các Cấu Trúc Hàng Đội Trong C++ STL

Container	Cơ chế	Thao tác chính	Đặc tính nổi bật
<code>std::stack</code>	LIFO	<code>push()</code> , <code>pop()</code> , <code>top()</code>	Truy xuất đỉnh $O(1)$, đóng gói quanh <code>deque</code> / <code>vector</code>
<code>std::queue</code>	FIFO	<code>push()</code> , <code>pop()</code> , <code>front()</code>	Vào trước ra trước, dùng trong BFS đồ thị
<code>std::deque</code>	Hai đầu	<code>push_front()</code> , <code>push_back()</code>	Chèn/xóa ở cả 2 đầu $O(1)$, chia thành nhiều chunk bộ nhớ
<code>std::priority_queue</code>	Heap	<code>push()</code> , <code>pop()</code> , <code>top()</code>	Cây vun đồng nhị phân, lấy cực trị $O(1)$, chèn/xóa $O(\log N)$

16.2 Kỹ Thuật Ngăn Xếp Đơn Điều (Monotonic Stack)

Xét bài toán kinh điển: **Next Greater Element (NGE)** - Với mỗi phần tử $A[i]$, tìm phần tử đầu tiên bên phải có giá trị lớn hơn $A[i]$.

Thuật toán ngây thơ: Với mỗi i , duyệt j từ $i + 1$ đến $N - 1$. Chi phí $O(N^2)$. **Thuật toán Ngăn Xếp Đơn Điều:** Duyệt mảng từ phải qua trái ($i = N - 1$ lùi về 0):

- Chùng nào ngăn xếp không rỗng và phần tử đỉnh ngăn xếp nhỏ hơn hoặc bằng $A[i]$: loại bỏ đỉnh (`pop()`). Vì $A[i]$ đứng trước và lớn hơn, đỉnh cũ sẽ không bao giờ có thể là NGE của bất kỳ phần tử nào đứng bên trái $A[i]$.
- Nếu ngăn xếp còn phần tử: đỉnh ngăn xếp chính là NGE của $A[i]$. Ngược lại, $A[i]$ không có NGE (ghi nhận -1).
- Đẩy $A[i]$ vào ngăn xếp.

Mỗi phần tử được `push` đúng 1 lần và `pop` tối đa 1 lần, tổng độ phức tạp toàn bộ mảng là $O(N)$!

[MỔ XẺ CÚ PHÁP C++14]	
<pre>for (int i = n - 1; i ≥ 0; --i) { while (!st.empty() && st.top() ≤ a[i]) st.pop(); nge[i] = st.empty() ? -1 : st.top(); st.push(a[i]); }</pre>	
Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<pre>for (int i = n - 1; i ≥ 0; --i)</pre>	Duyệt ngược từ phải sang trái để tích lũy các ứng viên bên phải.
<pre>while (!st.empty() && st.top() ≤ a[i]) st.pop();</pre>	Loại bỏ các phần tử bị che khuất bởi a[i] (duy trì đơn điệu giảm dần).
<pre>nge[i] = st.empty() ? -1 : st.top();</pre>	Nếu ngăn xếp còn phần tử, đỉnh chính là phần tử lớn hơn đầu tiên.
<pre>st.push(a[i]);</pre>	Đẩy a[i] vào làm ứng viên tiềm năng cho các phần tử bên trái nó.

16.3 Bảng Chạy Bàn Trên Giấy: Mảng [4, 5, 2, 25]

Bước i	Giá trị a[i]	Thao tác trên Stack	Stack sau bước	Kết quả nge[i]
Khởi tạo	-	Stack rỗng	[]	-
i=3	a[3]=25	Stack rỗng -> không pop	[25]	-1
i=2	a[2]=2	top=25 > 2 → giữ nguyên	[25, 2]	25
i=1	a[1]=5	top=2 ≤ 5 → pop(2); top=25 > 5	[25, 5]	25
i=0	a[0]=4	top=5 > 4 → giữ nguyên	[25, 5, 4]	5
Kết quả	-	-	-	*5 25 25 -1*

16.4 Bài Tập Tiêu Chuẩn

Bài toán: Bài 16.1 - Ngăn Xếp Đơn Điều: Phần Tử Lớn Hơn Đầu Tiên Bên Phải (Next Greater Element) <i>(Nguồn: Nền tảng C++14 - Ngăn xếp đơn điệu)</i>
Cho dãy N số nguyên dương $A_1, A_2, \dots, A_N$ ($1 \leq N \leq 15$). Với mỗi vị trí i ($1 \leq i \leq N$), hãy tìm phần tử đầu tiên nằm bên phải $A[i]$ có giá trị lớn hơn hẳn $A[i]$. Nếu không tồn tại phần tử nào lớn hơn, quy ước giá trị là -1 .

Đầu vào (Input): Dòng 1 ghi số nguyên N . Dòng 2 ghi N số nguyên $A_1, A_2, \dots, A_N$.

Đầu ra (Output): In ra N số nguyên là kết quả Next Greater Element tương ứng, cách nhau một khoảng trắng.

Ví dụ mẫu:

Đầu vào (Input):

4
 4 5 2 25

Đầu ra (Output):

5 25 25 -1

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): *Đáp án đối chiếu xem tại Phụ lục cuối sách*

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	[.....]	
2	Warm-up	[.....]	
3	Warm-up	[.....]	
4	Edge case	[.....]	
5	Edge case	[.....]	
6	Edge case	[.....]	
7	Trick case	[.....]	
8	Trick case	[.....]	
9	Mental challenge	[.....]	
10	Mental challenge	[.....]	

Chương 17: Tập Hợp & Ánh Xạ STL

[THUẬT NGỮ KHOA HỌC]

Cây Đỏ-Đen (Red-Black Tree) vs Bảng Băm (Hash Table)

Từ nguyên học:

Cây Đỏ-Đen do Rudolf Bayer phát minh năm 1972 dưới tên Cây B-nhi phân đối xứng và được hoàn thiện bởi Leonidas Guibas và Robert Sedgwick năm 1978.

Mô hình trực giác:

std::set và std::map duy trì các phần tử luôn được sắp xếp theo thứ tự nghiêm ngặt nhờ cây nhị phân tự cân bằng, đảm bảo mọi thao tác tìm kiếm, chèn, xóa đều chạy trong $O(\log N)$. Ngược lại, std::unordered_set và std::unordered_map phân tán dữ liệu vào các bucket thông qua hàm băm, đạt tốc độ $O(1)$ trung bình nhưng tiềm ẩn nguy cơ bị tấn công va chạm băm dẫn đến $O(N)$ trong thi đấu.

17.1 So Sánh Chi Tiết Giữa Hai Họ Cấu Trúc Dữ Liệu

Đặc tính	<code>std::set</code> <code>std::map</code>	<code>std::unordered_set</code> <code>std::unordered_map</code>
Cấu trúc dữ liệu	Cây Đỏ-Đen (Red-Black Tree)	Bảng băm (Hash Table)
Độ phức tạp (TB)	$O(\log N)$	$O(1)$
Trường hợp xấu nhất	$O(\log N)$	$O(N)$ (Khi bị va chạm băm)
Thứ tự phần tử	Luôn được sắp xếp tăng dần	Không có thứ tự xác định
Truy vấn khoảng	Hỗ trợ <code>lower_bound</code> , <code>upper_bound</code>	Không hỗ trợ
Mức độ an toàn trong CP	Tuyệt đối an toàn trước mọi test hiểm	Dễ bị hack test đối kháng (anti-hash tests)

17.2 Cạm Bẫy Sống Còn: `std::multiset::erase()`

[DANGER] Phân biệt xóa giá trị vs xóa con trỏ trong multiset

Nếu tập hợp `std::multiset<int> ms` chứa 5 phần tử có cùng giá trị 10:

- Lệnh `ms.erase(10);` sẽ xóa sạch toàn bộ cả 5 phần tử số 10!
- Để chỉ xóa duy nhất 1 phần tử số 10, bắt buộc phải tìm iterator của nó rồi xóa qua iterator:

```
auto it = ms.find(10);
if (it != ms.end()) ms.erase(it);
```

17.3 Mổ Xẻ Cú Pháp `lower_bound` Trên `std::set`

[MỔ XẺ CÚ PHÁP C++14]

```
std::set<int> s; auto it = s.lower_bound(x); if (it != s.end()) { int val = *it; }
```

Thành phần cú pháp	Cơ chế ngữ nghĩa & Mục đích
<code>s.lower_bound(x);</code>	LƯU Ý: Gọi phương thức thành viên <code>s.lower_bound()</code> $O(\log N)$. ĐỪNG BAO GIỜ gọi <code>std::lower_bound(s.begin(), s.end(), x)</code> vì sẽ mất $O(N)$!
<code>it != s.end()</code>	Kiểm tra xem trong tập hợp có tồn tại phần tử nào $\geq x$ hay không.
<code>int val = *it;</code>	Giải tham chiếu con trỏ iterator để lấy giá trị phần tử tìm được.

17.4 Bảng Chạy Bàn Trên Giấy: Chuỗi Truy Vấn Trên Cây Đỏ-Đen

Chuỗi thao tác: Chèn 10, Chèn 20, Chèn 10, Hỏi `lower_bound(15)`, Hỏi `lower_bound(25)`.

Truy vấn	Loại	Tập hợp Set	Bảng đếm Map	Kết quả in ra
1 10	Chèn 10	{10}	{10: 1}	-
1 20	Chèn 20	{10, 20}	{10: 1, 20: 1}	-
1 10	Chèn 10	{10, 20}	{10: 2, 20: 1}	-
2 15	<code>lower_bound(15)</code>	{10, 20}	-	20 (phần tử nhỏ nhất ≥ 15)
2 25	<code>lower_bound(25)</code>	{10, 20}	-	-1 (không có phần tử ≥ 25)

Truy vấn	Loại	Tập hợp Set	Bảng đếm Map	Kết quả in ra
Thống kê cuối	-	3 phần tử	Max freq: 10 (2 lần)	*Set size = 3, Max freq = 10*

17.5 Bài Tập Tiêu Chuẩn

Bài toán: Bài 17.1 - Tập Hợp & Ánh Xạ: Thống Kê Tần Suất & Truy Vấn Ngưỡng Cận

(Nguồn: Nền tảng C++14 - Cấu trúc dữ liệu STL Map & Set)

Cho N truy vấn thuộc 2 loại:

- Loại 1: `1 X` - Chèn giá trị nguyên X vào hệ thống (lưu trong `std::map<int, int>` đếm tần suất và `std::set<int>` lưu giá trị duy nhất).
- Loại 2: `2 X` - Tìm phần tử nhỏ nhất trong tập hợp có giá trị $\geq X$ (`lower_bound`). Nếu không có phần tử nào $\geq X$, in ra `-1`.

Sau khi kết thúc toàn bộ N truy vấn, in thêm trên dòng cuối: số lượng phần tử khác nhau duy nhất trong tập hợp `set_size` và phần tử có tần suất xuất hiện nhiều nhất `max_freq_val` (nếu hòa lấy phần tử nhỏ hơn).

Đầu vào (Input): Dòng 1 ghi N ($1 \leq N \leq 20$). N dòng tiếp theo mỗi dòng ghi một truy vấn dạng `1 X` hoặc `2 X`.

Đầu ra (Output): In kết quả từng truy vấn loại 2, và dòng cuối in `set_size max_freq_val`.

Ví dụ mẫu:

Đầu vào (Input):

```
6
1 10
1 20
1 10
2 15
2 25
1 30
```

Đầu ra (Output):

```
20
-1
3 10
```

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì): *Đáp án đối chiếu xem tại Phụ lục cuối sách*

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	<pre>6 1 10 1 20 1 10 2 15 2 25 1 30</pre>	

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
2	Warm-up	4 1 5 1 15 2 5 2 6	[.....]
3	Warm-up	4 1 100 2 50 2 100 2 150	[.....]
4	Edge case	2 2 10 2 20	[.....]
5	Edge case	3 1 7 1 7 1 7	[.....]
6	Edge case	4 1 3 1 8 2 3 2 8	[.....]
7	Trick case	3 1 20 1 10 2 15	[.....]
8	Trick case	4 1 -10 1 -5 2 -8 2 0	[.....]
9	Mental challenge	7 1 4 1 2 1 6 2 3 1 2 2 5 2 7	[.....]
10	Mental challenge	7 1 1 1 5 1 3 1 5 2 2 2 4 1 3	[.....]

Bài Tập Tích Hợp Tập 2: Phân Đoạn & Ghép Cặp Tài Nguyên Tối Ưu

[THUẬT NGỮ KHOA HỌC] Hệ Thống Phối Hợp Giải Thuật Đa Tầng

- Từ nguyên học:** Trong các kỳ thi HSG Quốc gia, Olympic Tin học và các kỳ thi Competitive Programming quốc tế (ICPC, Codeforces Div 1/2), bài toán hiếm khi đứng độc lập dưới một thuật toán đơn lẻ. Sức mạnh thực sự nằm ở năng lực kết hợp nhịp nhàng giữa Cấu trúc Dữ liệu STL, Kỹ thuật Hai Con Trỏ và Thuật toán Chặt Nhị Phân.
- Mô hình trực giác:** Một bài toán hoàn chỉnh được giải quyết qua 3 tầng: Tầng 1 thu thập thống kê cấu trúc bằng Set & Map, Tầng 2 khai thác tính đơn điệu để đếm cặp bằng Hai Con Trỏ, và Tầng 3 tìm ngưỡng tối ưu toàn cục bằng Chặt nhị phân kết quả.

Kiến Trúc 3 Công Đoạn Giải Thuật

1. Công đoạn 1 (Lọc Dữ Liệu & Tần Suất Bằng STL Set & Map):

- `std::set<long long>` loại bỏ các giá trị trùng lặp và xác định số phần tử độc nhất `unique_count`.
- `std::map<long long, int>` đếm số lần xuất hiện của từng phần tử để tìm `max_freq_val` (giá trị có tần suất cao nhất, ưu tiên giá trị nhỏ hơn khi hòa).

2. Công đoạn 2 (Đếm Cặp Bằng Hai Con Trỏ Trên Mảng Sắp Xếp):

- Sắp xếp một bản sao mảng B tăng dần.
- Đặt hai con trỏ $L = 0$ và $R = N - 1$.
- Nếu $B[L] + B[R] \leq S$, thì $B[L]$ kết hợp với mọi phần tử $B[j]$ với $j \in [L + 1, R]$ đều có tổng $\leq S$. Ta cộng ngay $(R - L)$ cặp và tăng $L++$.
- Nếu $B[L] + B[R] > S$, giảm $R--$.

3. Công đoạn 3 (Chặt Nhị Phân Kết Quả Phân Đoạn Công Bằng):

- Chia mảng ban đầu A (giữ nguyên thứ tự tuần tự) cho K cụm máy sao cho tải trọng cực đại của một cụm là nhỏ nhất có thể.
- Không gian tìm kiếm: $L = \max(A)$, $R = \sum A$. Hàm `check(mid)` đếm số đoạn con liên tiếp có tổng $\leq M$.

Bảng Chạy Bàn Tổng Hợp Cho Test Mẫu [3, 1, 4, 1, 5] Với $S = 7, K = 2$

Công đoạn	Phương pháp	Dữ liệu xử lý	Trạng thái	Kết quả đầu ra
Tầng 1	STL Set & Map	[3, 1, 4, 1, 5]	Set={1, 3, 4, 5}; Freq={1:2, 3:1, 4:1, 5:1}	Uniq = 4, Best = 1

Công đoạn	Phương pháp	Dữ liệu xử lý	Trạng thái	Kết quả đầu ra
Tầng 2	Hai Con Trỏ	<code>B=[1, 1, 3, 4, 5], S=7</code>	<code>L=0 (+4), L=1 (+3), L=2 (+1)</code>	<code>Pair count = 8</code>
Tầng 3	Chặt Nhị Phân	<code>A=[3, 1, 4, 1, 5], K=2</code>	<code>mid=8: [3, 1, 4] và [1, 5] (2 đoạn ≤ 2)</code>	<code>Min max workload = 8</code>
Kết quả tổng	Ghép nối 3 tầng	-	-	<code>*4 1 8 8*</code>

Đề Bài Chi Tiết

Bài toán: Bài Tập Tích Hợp 2 - Phân Đoạn & Ghép Cặp Tài Nguyên Tối Ưu

(Nguồn: Chuyên khảo CP C++14 - Tích hợp STL & Kỹ thuật Cốt lõi)

Một trung tâm điện toán đám mây nhận được danh sách N gói tài nguyên có kích thước $A_1, A_2, \dots, A_N$. Hệ thống cần thực hiện 3 công đoạn tối ưu hóa:

- Đếm số loại kích thước duy nhất (`unique_count`) và tìm kích thước có tần suất lớn nhất (`max_freq_val`) (nếu hòa tần suất, chọn kích thước nhỏ hơn).
- Sau khi sắp xếp dãy kích thước theo thứ tự tăng dần, sử dụng thuật toán Hai Con Trỏ để đếm số cặp (i, j) ($i < j$) sao cho tổng kích thước $A_i + A_j \leq S$ (`pair_count`).
- Chia N gói tài nguyên theo thứ tự ban đầu cho K cụm máy chủ sao cho tải trọng lớn nhất của một cụm là nhỏ nhất có thể (`min_max_workload`).

Đầu vào (Input): Dòng 1 ghi ba số nguyên N, S, K ($1 \leq N \leq 15, 1 \leq S \leq 10^9, 1 \leq K \leq N$). Dòng 2 ghi N số nguyên dương $A_1, A_2, \dots, A_N$ ($1 \leq A_i \leq 10^9$).

Đầu ra (Output): In ra trên một dòng 4 số nguyên cách nhau một khoảng trắng: `unique_count max_freq_val pair_count min_max_workload`.

Ví dụ mẫu:

Đầu vào (Input):

Đầu ra (Output):

5 7 2
3 1 4 1 5

4 1 8 8

Thử thách 10 Test Bàn Tay (Tính nhẩm & Điền bằng bút chì):

Đáp án đối chiếu xem tại Phụ lục cuối sách

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
1	Warm-up	5 7 2 3 1 4 1 5	
2	Warm-up	4 10 2 2 4 6 8	

Test	Phân loại	Dữ liệu vào (Input)	Đáp án của bạn (Bút chì)
3	Warm-up	5 6 3 1 2 3 2 1	
4	Edge case	1 10 1 5	
5	Edge case	4 8 2 4 4 4 4	
6	Edge case	4 20 4 10 20 30 40	
7	Trick case	4 3 2 5 6 7 8	
8	Trick case	3 15 1 2 5 3	
9	Mental challenge	6 10 3 1 5 2 8 3 4	
10	Mental challenge	6 8 2 2 3 2 4 1 3	

Phụ Lục: Đáp Án & Bảng Vết Thực Thi 10 Test Bàn Tay

[LÝ THUYẾT] Hướng Dẫn Đối Chiếu Lời Giải Bằng Bút Chì
Phụ lục này cung cấp toàn bộ đáp án chính thức và bảng vết thực thi (Execution Trace) từng bước cho tất cả 10 bài test của từng chương trong Tập 2. Sau khi đã tự giải và điền kết quả vào các ô trống bằng bút chì, các em hãy đối chiếu với bảng đáp án dưới đây để phát hiện chính xác bước tư duy nào bị nhầm lẫn.

Lời Giải Chi Tiết Chương 9: Số Học Cơ Bản Cho CP

Lời giải & Đáp án: Bài 9.1 - Số Học Cơ Bản Cho CP			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	3 5 7 10	1 5 4	Tính nhẩm cơ bản số nhỏ
2	12 18 100 20	6 4 8	Ước chung và lũy thừa chẵn
3	4 6 10 5	2 6 3	Bộ số nhỏ với modulo 10
4	1 1 13 1	1 1 0	Cận dưới A=1, B=1, K=1 không có nguyên tố
5	7 7 7 2	7 0 1	A bằng B bằng M, modulo bằng 0
6	15 1 1000 1	1 15 0	Số mũ B=1 giữ nguyên cơ số
7	10 3 5 15	1 0 6	Cơ số chia hết cho modulo
8	2 10 10000000007 25	2 1024 9	Modulo lớn 10^9+7 với lũy thừa 2^{10}
9	6 8 11 8	2 4 4	Tính nhẩm lũy thừa modulo 11 và số nguyên tố ≤ 8
10	5 4 13 7	1 1 4	Tính nhẩm $5^4 \bmod 13$ qua $5^2 = -1$

Bảng Vết Thực Thi Từng Bước (Chương 9)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	3
-	b	5
-	m	7
-	k	10
11	Euclid step: a % b	-
-	a	3
-	b	5
-	r	3
12	Euclid step: a % b	-
-	a	5

Bước	Mô tả / Biến	Giá trị
-	b	3
-	r	2
13	Euclid step: $a \% b$	-
-	a	3
-	b	2
-	r	1
14	Euclid step: $a \% b$	-
-	a	2
-	b	1
-	r	0
2	GCD calculated	-
-	gcd	1
21	Exp odd: multiply res with base	-
-	res	3
31	Square base and shift exp	-
-	base	2
-	exp	2
32	Square base and shift exp	-
-	base	4
-	exp	1
23	Exp odd: multiply res with base	-
-	res	5
33	Square base and shift exp	-
-	base	2
-	exp	0
3	Power Mod calculated	-
-	pow_mod	5
4	Primes counted	-
-	primes_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	12
-	b	18
-	m	100
-	k	20
11	Euclid step: $a \% b$	-
-	a	12
-	b	18
-	r	12
12	Euclid step: $a \% b$	-

Bước	Mô tả / Biến	Giá trị
-	a	18
-	b	12
-	r	6
13	Euclid step: $a \% b$	-
-	a	12
-	b	6
-	r	0
2	GCD calculated	-
-	gcd	6
31	Square base and shift exp	-
-	base	44
-	exp	9
22	Exp odd: multiply res with base	-
-	res	44
32	Square base and shift exp	-
-	base	36
-	exp	4
33	Square base and shift exp	-
-	base	96
-	exp	2
34	Square base and shift exp	-
-	base	16
-	exp	1
25	Exp odd: multiply res with base	-
-	res	4
35	Square base and shift exp	-
-	base	56
-	exp	0
3	Power Mod calculated	-
-	pow_mod	4
4	Primes counted	-
-	primes_count	8

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	4
-	b	6
-	m	10
-	k	5
11	Euclid step: $a \% b$	-
-	a	4

Bước	Mô tả / Biến	Giá trị
-	b	6
-	r	4
12	Euclid step: $a \% b$	-
-	a	6
-	b	4
-	r	2
13	Euclid step: $a \% b$	-
-	a	4
-	b	2
-	r	0
2	GCD calculated	-
-	gcd	2
31	Square base and shift exp	-
-	base	6
-	exp	3
22	Exp odd: multiply res with base	-
-	res	6
32	Square base and shift exp	-
-	base	6
-	exp	1
23	Exp odd: multiply res with base	-
-	res	6
33	Square base and shift exp	-
-	base	6
-	exp	0
3	Power Mod calculated	-
-	pow_mod	6
4	Primes counted	-
-	primes_count	3

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	1
-	b	1
-	m	13
-	k	1
11	Euclid step: $a \% b$	-
-	a	1
-	b	1
-	r	0
2	GCD calculated	-

Bước	Mô tả / Biến	Giá trị
-	gcd	1
21	Exp odd: multiply res with base	-
-	res	1
31	Square base and shift exp	-
-	base	1
-	exp	0
3	Power Mod calculated	-
-	pow_mod	1
4	Primes counted	-
-	primes_count	0

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	7
-	b	7
-	m	7
-	k	2
11	Euclid step: $a \% b$	-
-	a	7
-	b	7
-	r	0
2	GCD calculated	-
-	gcd	7
21	Exp odd: multiply res with base	-
-	res	0
31	Square base and shift exp	-
-	base	0
-	exp	3
22	Exp odd: multiply res with base	-
-	res	0
32	Square base and shift exp	-
-	base	0
-	exp	1
23	Exp odd: multiply res with base	-
-	res	0
33	Square base and shift exp	-
-	base	0
-	exp	0
3	Power Mod calculated	-
-	pow_mod	0
4	Primes counted	-

Bước	Mô tả / Biến	Giá trị
-	primes_count	1

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	15
-	b	1
-	m	1000
-	k	1
11	Euclid step: $a \% b$	-
-	a	15
-	b	1
-	r	0
2	GCD calculated	-
-	gcd	1
21	Exp odd: multiply res with base	-
-	res	15
31	Square base and shift exp	-
-	base	225
-	exp	0
3	Power Mod calculated	-
-	pow_mod	15
4	Primes counted	-
-	primes_count	0

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	10
-	b	3
-	m	5
-	k	15
11	Euclid step: $a \% b$	-
-	a	10
-	b	3
-	r	1
12	Euclid step: $a \% b$	-
-	a	3
-	b	1
-	r	0
2	GCD calculated	-
-	gcd	1
21	Exp odd: multiply res with base	-

Bước	Mô tả / Biến	Giá trị
-	res	0
31	Square base and shift exp	-
-	base	0
-	exp	1
22	Exp odd: multiply res with base	-
-	res	0
32	Square base and shift exp	-
-	base	0
-	exp	0
3	Power Mod calculated	-
-	pow_mod	0
4	Primes counted	-
-	primes_count	6

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	2
-	b	10
-	m	1000000007
-	k	25
11	Euclid step: $a \% b$	-
-	a	2
-	b	10
-	r	2
12	Euclid step: $a \% b$	-
-	a	10
-	b	2
-	r	0
2	GCD calculated	-
-	gcd	2
31	Square base and shift exp	-
-	base	4
-	exp	5
22	Exp odd: multiply res with base	-
-	res	4
32	Square base and shift exp	-
-	base	16
-	exp	2
33	Square base and shift exp	-
-	base	256
-	exp	1

Bước	Mô tả / Biến	Giá trị
24	Exp odd: multiply res with base	-
-	res	1024
34	Square base and shift exp	-
-	base	65536
-	exp	0
3	Power Mod calculated	-
-	pow_mod	1024
4	Primes counted	-
-	primes_count	9

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	6
-	b	8
-	m	11
-	k	8
11	Euclid step: $a \% b$	-
-	a	6
-	b	8
-	r	6
12	Euclid step: $a \% b$	-
-	a	8
-	b	6
-	r	2
13	Euclid step: $a \% b$	-
-	a	6
-	b	2
-	r	0
2	GCD calculated	-
-	gcd	2
31	Square base and shift exp	-
-	base	3
-	exp	4
32	Square base and shift exp	-
-	base	9
-	exp	2
33	Square base and shift exp	-
-	base	4
-	exp	1
24	Exp odd: multiply res with base	-
-	res	4

Bước	Mô tả / Biến	Giá trị
34	Square base and shift exp	-
-	base	5
-	exp	0
3	Power Mod calculated	-
-	pow_mod	4
4	Primes counted	-
-	primes_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	a	5
-	b	4
-	m	13
-	k	7
11	Euclid step: $a \% b$	-
-	a	5
-	b	4
-	r	1
12	Euclid step: $a \% b$	-
-	a	4
-	b	1
-	r	0
2	GCD calculated	-
-	gcd	1
31	Square base and shift exp	-
-	base	12
-	exp	2
32	Square base and shift exp	-
-	base	1
-	exp	1
23	Exp odd: multiply res with base	-
-	res	1
33	Square base and shift exp	-
-	base	1
-	exp	0
3	Power Mod calculated	-
-	pow_mod	1
4	Primes counted	-
-	primes_count	4

Lời Giải Chi Tiết Chương 10: Cấu Trúc Dữ Liệu Tuần Tự STL

Lời giải & Đáp án: Bài 10.1 - Điểm Tọa Độ Vector & Chi Phí Khấu Hao			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	4 1 2 3 4 -5 -2 2 1	2 1 2	Thao tác push và pop cơ bản
2	3 5 5 2 8 -1 4	3 -1 8	Ba thao tác push liên tiếp với tọa độ âm
3	2 0 0 4 -2	2 0 0	Tổng tọa độ bằng 0 và dương
4	1 -3 5	1 -3 5	Một phần tử duy nhất thỏa mãn push
5	1 -2 -3	0 0 0	Thao tác pop khi vector rỗng
6	4 1 1 2 2 -3 -1 -4 -4	0 0 0	Đẩy 2 phần tử rồi pop hết trở về rỗng
7	4 -1 -1 -2 -2 3 3 4 4	2 3 4	Pop liên tiếp khi rỗng rồi mới push
8	5 1 0 -1 -2 2 0 -3 -3 5 5	1 5 5	Thao tác xen kẽ push và pop liên tục
9	6 2 3 1 5 -4 -1 3 1 0 2 -5 -2	2 2 3	Theo dõi trạng thái 6 điểm trên giấy
10	5 -2 2 4 -1 -1 7 -6 2 3 0	3 -2 2	Biên tổng bằng 0 và tìm min max cuối cùng

Bảng Vết Thực Thi Từng Bước (Chương 10)

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	4
10	Process input pair (x, y)	-
-	x	1
-	y	2
-	sum	3
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	3
-	y	4
-	sum	7
101	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
12	Process input pair (x, y)	-
-	x	-5
-	y	-2
-	sum	-7
202	Pop back last pair from vector	-
-	current_size	1
-	current_capacity	2
13	Process input pair (x, y)	-
-	x	2
-	y	1
-	sum	3
103	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
999	Calculated min_x and max_y	-
-	min_x	1
-	max_y	2

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	3
10	Process input pair (x, y)	-
-	x	5

Bước	Mô tả / Biến	Giá trị
-	y	5
-	sum	10
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	2
-	y	8
-	sum	10
101	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
12	Process input pair (x, y)	-
-	x	-1
-	y	4
-	sum	3
102	Push back pair into vector	-
-	current_size	3
-	current_capacity	4
999	Calculated min_x and max_y	-
-	min_x	-1
-	max_y	8

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	2
10	Process input pair (x, y)	-
-	x	0
-	y	0
-	sum	0
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	4
-	y	-2
-	sum	2
101	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
999	Calculated min_x and max_y	-

Bước	Mô tả / Biến	Giá trị
-	min_x	0
-	max_y	0

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	1
10	Process input pair (x, y)	-
-	x	-3
-	y	5
-	sum	2
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
999	Calculated min_x and max_y	-
-	min_x	-3
-	max_y	5

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	1
10	Process input pair (x, y)	-
-	x	-2
-	y	-3
-	sum	-5
300	Vector already empty, ignore pop	-
-	current_size	0
-	current_capacity	0
999	Vector is empty	-

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	4
10	Process input pair (x, y)	-
-	x	1
-	y	1
-	sum	2
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	2
-	y	2

Bước	Mô tả / Biến	Giá trị
-	sum	4
101	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
12	Process input pair (x, y)	-
-	x	-3
-	y	-1
-	sum	-4
202	Pop back last pair from vector	-
-	current_size	1
-	current_capacity	2
13	Process input pair (x, y)	-
-	x	-4
-	y	-4
-	sum	-8
203	Pop back last pair from vector	-
-	current_size	0
-	current_capacity	2
999	Vector is empty	-

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	4
10	Process input pair (x, y)	-
-	x	-1
-	y	-1
-	sum	-2
300	Vector already empty, ignore pop	-
-	current_size	0
-	current_capacity	0
11	Process input pair (x, y)	-
-	x	-2
-	y	-2
-	sum	-4
301	Vector already empty, ignore pop	-
-	current_size	0
-	current_capacity	0
12	Process input pair (x, y)	-
-	x	3
-	y	3
-	sum	6

Bước	Mô tả / Biến	Giá trị
102	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
13	Process input pair (x, y)	-
-	x	4
-	y	4
-	sum	8
103	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
999	Calculated min_x and max_y	-
-	min_x	3
-	max_y	4

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	5
10	Process input pair (x, y)	-
-	x	1
-	y	0
-	sum	1
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	-1
-	y	-2
-	sum	-3
201	Pop back last pair from vector	-
-	current_size	0
-	current_capacity	1
12	Process input pair (x, y)	-
-	x	2
-	y	0
-	sum	2
102	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
13	Process input pair (x, y)	-
-	x	-3
-	y	-3

Bước	Mô tả / Biến	Giá trị
-	sum	-6
203	Pop back last pair from vector	-
-	current_size	0
-	current_capacity	1
14	Process input pair (x, y)	-
-	x	5
-	y	5
-	sum	10
104	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
999	Calculated min_x and max_y	-
-	min_x	5
-	max_y	5

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	6
10	Process input pair (x, y)	-
-	x	2
-	y	3
-	sum	5
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	1
-	y	5
-	sum	6
101	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
12	Process input pair (x, y)	-
-	x	-4
-	y	-1
-	sum	-5
202	Pop back last pair from vector	-
-	current_size	1
-	current_capacity	2
13	Process input pair (x, y)	-
-	x	3

Bước	Mô tả / Biến	Giá trị
-	y	1
-	sum	4
103	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
14	Process input pair (x, y)	-
-	x	0
-	y	2
-	sum	2
104	Push back pair into vector	-
-	current_size	3
-	current_capacity	4
15	Process input pair (x, y)	-
-	x	-5
-	y	-2
-	sum	-7
205	Pop back last pair from vector	-
-	current_size	2
-	current_capacity	4
999	Calculated min_x and max_y	-
-	min_x	2
-	max_y	3

Bước	Mô tả / Biến	Giá trị
1	Initialize empty vector of pairs	-
-	n	5
10	Process input pair (x, y)	-
-	x	-2
-	y	2
-	sum	0
100	Push back pair into vector	-
-	current_size	1
-	current_capacity	1
11	Process input pair (x, y)	-
-	x	4
-	y	-1
-	sum	3
101	Push back pair into vector	-
-	current_size	2
-	current_capacity	2
12	Process input pair (x, y)	-

Bước	Mô tả / Biến	Giá trị
-	x	-1
-	y	7
-	sum	6
102	Push back pair into vector	-
-	current_size	3
-	current_capacity	4
13	Process input pair (x, y)	-
-	x	-6
-	y	2
-	sum	-4
203	Pop back last pair from vector	-
-	current_size	2
-	current_capacity	4
14	Process input pair (x, y)	-
-	x	3
-	y	0
-	sum	3
104	Push back pair into vector	-
-	current_size	3
-	current_capacity	4
999	Calculated min_x and max_y	-
-	min_x	-2
-	max_y	2

Lời Giải Chi Tiết Chương 11: Kỹ Thuật Hai Con Trỏ & Cửa Sổ Trượt

Lời giải & Đáp án: Bài 11.1 - Cửa Sổ Trượt: Đoạn Con Liên Tiếp Có Tổng Không Vượt Quá S			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	5 7 2 1 3 4 1	3 10	Cửa sổ trượt cơ bản với mảng 5 phần tử
2	4 5 1 2 1 1	4 10	Toàn bộ mảng có tổng <= S
3	4 3 1 1 1 1	3 9	Mảng các số 1 với giới hạn S=3
4	1 10 5	1 1	Mảng 1 phần tử nhỏ hơn S
5	1 5 10	0 0	Mảng 1 phần tử lớn hơn S
6	3 2 5 7 9	0 0	Mọi phần tử đều vượt quá ngưỡng S
7	5 4 2 10 3 1 10	2 4	Phần tử cực lớn ngắt quãng cửa sổ
8	3 6 1 2 3	3 6	Tổng toàn mảng đúng bằng S
9	6 6 3 2 1 4 2 1	3 12	Theo dõi hai con trỏ co giãn qua 6 bước
10	6 5 2 2 2 2 2 2	2 11	Mảng đồng nhất với cửa sổ cực đại độ dài 2

Bảng Vết Thực Thi Từng Bước (Chương 11)

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	5
-	s	7
-	a	(2, 1, 3, 4, 1)
10	Expand right boundary	-
-	right	0
-	a[right]	2
-	current_sum	2
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	1
-	current_sum	3
-	valid_window_len	2

Bước	Mô tả / Biến	Giá trị
-	subarrays_ending_at_right	2
12	Expand right boundary	-
-	right	2
-	a[right]	3
-	current_sum	6
-	valid_window_len	3
-	subarrays_ending_at_right	3
13	Expand right boundary	-
-	right	3
-	a[right]	4
-	current_sum	10
100	Shrink left boundary	-
-	left_removed	2
-	new_left	1
-	current_sum	8
101	Shrink left boundary	-
-	left_removed	1
-	new_left	2
-	current_sum	7
-	valid_window_len	2
-	subarrays_ending_at_right	2
14	Expand right boundary	-
-	right	4
-	a[right]	1
-	current_sum	8
102	Shrink left boundary	-
-	left_removed	3
-	new_left	3
-	current_sum	5
-	valid_window_len	2
-	subarrays_ending_at_right	2
999	Finished sliding window	-
-	max_len	3
-	total_count	10

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	4
-	s	5
-	a	(1, 2, 1, 1)
10	Expand right boundary	-

Bước	Mô tả / Biến	Giá trị
-	right	0
-	a[right]	1
-	current_sum	1
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	2
-	current_sum	3
-	valid_window_len	2
-	subarrays_ending_at_right	2
12	Expand right boundary	-
-	right	2
-	a[right]	1
-	current_sum	4
-	valid_window_len	3
-	subarrays_ending_at_right	3
13	Expand right boundary	-
-	right	3
-	a[right]	1
-	current_sum	5
-	valid_window_len	4
-	subarrays_ending_at_right	4
999	Finished sliding window	-
-	max_len	4
-	total_count	10

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	4
-	s	3
-	a	(1, 1, 1, 1)
10	Expand right boundary	-
-	right	0
-	a[right]	1
-	current_sum	1
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	1

Bước	Mô tả / Biến	Giá trị
-	current_sum	2
-	valid_window_len	2
-	subarrays_ending_at_right	2
12	Expand right boundary	-
-	right	2
-	a[right]	1
-	current_sum	3
-	valid_window_len	3
-	subarrays_ending_at_right	3
13	Expand right boundary	-
-	right	3
-	a[right]	1
-	current_sum	4
100	Shrink left boundary	-
-	left_removed	1
-	new_left	1
-	current_sum	3
-	valid_window_len	3
-	subarrays_ending_at_right	3
999	Finished sliding window	-
-	max_len	3
-	total_count	9

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	1
-	s	10
-	a	(5,)
10	Expand right boundary	-
-	right	0
-	a[right]	5
-	current_sum	5
-	valid_window_len	1
-	subarrays_ending_at_right	1
999	Finished sliding window	-
-	max_len	1
-	total_count	1

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	1

Bước	Mô tả / Biến	Giá trị
-	s	5
-	a	(10,)
10	Expand right boundary	-
-	right	0
-	a[right]	10
-	current_sum	10
100	Shrink left boundary	-
-	left_removed	10
-	new_left	1
-	current_sum	0
999	Finished sliding window	-
-	max_len	0
-	total_count	0

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	3
-	s	2
-	a	(5, 7, 9)
10	Expand right boundary	-
-	right	0
-	a[right]	5
-	current_sum	5
100	Shrink left boundary	-
-	left_removed	5
-	new_left	1
-	current_sum	0
11	Expand right boundary	-
-	right	1
-	a[right]	7
-	current_sum	7
101	Shrink left boundary	-
-	left_removed	7
-	new_left	2
-	current_sum	0
12	Expand right boundary	-
-	right	2
-	a[right]	9
-	current_sum	9
102	Shrink left boundary	-
-	left_removed	9

Bước	Mô tả / Biến	Giá trị
-	new_left	3
-	current_sum	0
999	Finished sliding window	-
-	max_len	0
-	total_count	0

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	5
-	s	4
-	a	(2, 10, 3, 1, 10)
10	Expand right boundary	-
-	right	0
-	a[right]	2
-	current_sum	2
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	10
-	current_sum	12
100	Shrink left boundary	-
-	left_removed	2
-	new_left	1
-	current_sum	10
101	Shrink left boundary	-
-	left_removed	10
-	new_left	2
-	current_sum	0
12	Expand right boundary	-
-	right	2
-	a[right]	3
-	current_sum	3
-	valid_window_len	1
-	subarrays_ending_at_right	1
13	Expand right boundary	-
-	right	3
-	a[right]	1
-	current_sum	4
-	valid_window_len	2
-	subarrays_ending_at_right	2

Bước	Mô tả / Biến	Giá trị
14	Expand right boundary	-
-	right	4
-	a[right]	10
-	current_sum	14
102	Shrink left boundary	-
-	left_removed	3
-	new_left	3
-	current_sum	11
103	Shrink left boundary	-
-	left_removed	1
-	new_left	4
-	current_sum	10
104	Shrink left boundary	-
-	left_removed	10
-	new_left	5
-	current_sum	0
999	Finished sliding window	-
-	max_len	2
-	total_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	3
-	s	6
-	a	(1, 2, 3)
10	Expand right boundary	-
-	right	0
-	a[right]	1
-	current_sum	1
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	2
-	current_sum	3
-	valid_window_len	2
-	subarrays_ending_at_right	2
12	Expand right boundary	-
-	right	2
-	a[right]	3
-	current_sum	6

Bước	Mô tả / Biến	Giá trị
-	valid_window_len	3
-	subarrays_ending_at_right	3
999	Finished sliding window	-
-	max_len	3
-	total_count	6

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	6
-	s	6
-	a	(3, 2, 1, 4, 2, 1)
10	Expand right boundary	-
-	right	0
-	a[right]	3
-	current_sum	3
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	2
-	current_sum	5
-	valid_window_len	2
-	subarrays_ending_at_right	2
12	Expand right boundary	-
-	right	2
-	a[right]	1
-	current_sum	6
-	valid_window_len	3
-	subarrays_ending_at_right	3
13	Expand right boundary	-
-	right	3
-	a[right]	4
-	current_sum	10
100	Shrink left boundary	-
-	left_removed	3
-	new_left	1
-	current_sum	7
101	Shrink left boundary	-
-	left_removed	2
-	new_left	2
-	current_sum	5

Bước	Mô tả / Biến	Giá trị
-	valid_window_len	2
-	subarrays_ending_at_right	2
14	Expand right boundary	-
-	right	4
-	a[right]	2
-	current_sum	7
102	Shrink left boundary	-
-	left_removed	1
-	new_left	3
-	current_sum	6
-	valid_window_len	2
-	subarrays_ending_at_right	2
15	Expand right boundary	-
-	right	5
-	a[right]	1
-	current_sum	7
103	Shrink left boundary	-
-	left_removed	4
-	new_left	4
-	current_sum	3
-	valid_window_len	2
-	subarrays_ending_at_right	2
999	Finished sliding window	-
-	max_len	3
-	total_count	12

Bước	Mô tả / Biến	Giá trị
1	Read inputs n, s and array a	-
-	n	6
-	s	5
-	a	(2, 2, 2, 2, 2, 2)
10	Expand right boundary	-
-	right	0
-	a[right]	2
-	current_sum	2
-	valid_window_len	1
-	subarrays_ending_at_right	1
11	Expand right boundary	-
-	right	1
-	a[right]	2
-	current_sum	4

Bước	Mô tả / Biến	Giá trị
-	valid_window_len	2
-	subarrays_ending_at_right	2
12	Expand right boundary	-
-	right	2
-	a[right]	2
-	current_sum	6
100	Shrink left boundary	-
-	left_removed	2
-	new_left	1
-	current_sum	4
-	valid_window_len	2
-	subarrays_ending_at_right	2
13	Expand right boundary	-
-	right	3
-	a[right]	2
-	current_sum	6
101	Shrink left boundary	-
-	left_removed	2
-	new_left	2
-	current_sum	4
-	valid_window_len	2
-	subarrays_ending_at_right	2
14	Expand right boundary	-
-	right	4
-	a[right]	2
-	current_sum	6
102	Shrink left boundary	-
-	left_removed	2
-	new_left	3
-	current_sum	4
-	valid_window_len	2
-	subarrays_ending_at_right	2
15	Expand right boundary	-
-	right	5
-	a[right]	2
-	current_sum	6
103	Shrink left boundary	-
-	left_removed	2
-	new_left	4
-	current_sum	4
-	valid_window_len	2

Bước	Mô tả / Biến	Giá trị
-	subarrays_ending_at_right	2
999	Finished sliding window	-
-	max_len	2
-	total_count	11

Lời Giải Chi Tiết Chương 12: Sắp Xếp & Tìm Kiếm Nhị Phân, Chặt Nhị Phân Kết Quả

Lời giải & Đáp án: Bài 12.1 - Chặt Nhị Phân Phân Đoạn Công Việc			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	5 3 2 4 7 1 6	7 3	Chia 5 việc cho 3 người với tải trọng cực tiểu 7
2	4 2 10 20 30 40	60 6	Chia 4 việc cho 2 người, tối ưu là [10, 20, 30] và [40]
3	3 3 5 10 15	15 4	Số người bằng số việc, tải trọng bằng max phần tử
4	1 1 42	42 1	Duy nhất một phần tử
5	5 1 1 2 3 4 5	15 4	Chỉ có 1 người làm toàn bộ công việc
6	4 4 7 7 7 7	7 4	Các công việc có độ dài bằng nhau hoàn toàn
7	5 2 100 1 1 1 1	100 2	Một phần tử lớn át toàn bộ mảng
8	6 3 1 1 1 1 1 1	2 3	Mảng toàn số 1 chia làm 3 đoạn bằng 2
9	5 2 3 2 4 1 5	9 4	Tính nhẩm chia [3, 2, 4] và [1, 5] trên giấy
10	6 2 1 2 3 4 5 6	11 4	Chia dãy tăng dần 1..6 thành 2 đoạn tối ưu

Bảng Vết Thực Thi Từng Bước (Chương 12)

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	5
-	k	3
-	a	(2, 4, 7, 1, 6)
-	low_initial	7
-	high_initial	20
11	Binary search iteration	-
-	low	7
-	high	20
-	mid	13
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	12
-	best_ans	13

Bước	Mô tả / Biến	Giá trị
12	Binary search iteration	-
-	low	7
-	high	12
-	mid	9
-	check_ok	true
102	mid feasible, search lower half	-
-	new_high	8
-	best_ans	9
13	Binary search iteration	-
-	low	7
-	high	8
-	mid	7
-	check_ok	true
103	mid feasible, search lower half	-
-	new_high	6
-	best_ans	7
999	Binary search finished	-
-	ans	7
-	check_count	3

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	4
-	k	2
-	a	(10, 20, 30, 40)
-	low_initial	40
-	high_initial	100
11	Binary search iteration	-
-	low	40
-	high	100
-	mid	70
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	69
-	best_ans	70
12	Binary search iteration	-
-	low	40
-	high	69
-	mid	54
-	check_ok	false
202	mid infeasible, search upper half	-

Bước	Mô tả / Biến	Giá trị
-	new_low	55
13	Binary search iteration	-
-	low	55
-	high	69
-	mid	62
-	check_ok	true
103	mid feasible, search lower half	-
-	new_high	61
-	best_ans	62
14	Binary search iteration	-
-	low	55
-	high	61
-	mid	58
-	check_ok	false
204	mid infeasible, search upper half	-
-	new_low	59
15	Binary search iteration	-
-	low	59
-	high	61
-	mid	60
-	check_ok	true
105	mid feasible, search lower half	-
-	new_high	59
-	best_ans	60
16	Binary search iteration	-
-	low	59
-	high	59
-	mid	59
-	check_ok	false
206	mid infeasible, search upper half	-
-	new_low	60
999	Binary search finished	-
-	ans	60
-	check_count	6

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	3
-	k	3
-	a	(5, 10, 15)
-	low_initial	15

Bước	Mô tả / Biến	Giá trị
-	high_initial	30
11	Binary search iteration	-
-	low	15
-	high	30
-	mid	22
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	21
-	best_ans	22
12	Binary search iteration	-
-	low	15
-	high	21
-	mid	18
-	check_ok	true
102	mid feasible, search lower half	-
-	new_high	17
-	best_ans	18
13	Binary search iteration	-
-	low	15
-	high	17
-	mid	16
-	check_ok	true
103	mid feasible, search lower half	-
-	new_high	15
-	best_ans	16
14	Binary search iteration	-
-	low	15
-	high	15
-	mid	15
-	check_ok	true
104	mid feasible, search lower half	-
-	new_high	14
-	best_ans	15
999	Binary search finished	-
-	ans	15
-	check_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	1
-	k	1

Bước	Mô tả / Biến	Giá trị
-	a	(42,)
-	low_initial	42
-	high_initial	42
11	Binary search iteration	-
-	low	42
-	high	42
-	mid	42
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	41
-	best_ans	42
999	Binary search finished	-
-	ans	42
-	check_count	1

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	5
-	k	1
-	a	(1, 2, 3, 4, 5)
-	low_initial	5
-	high_initial	15
11	Binary search iteration	-
-	low	5
-	high	15
-	mid	10
-	check_ok	false
201	mid infeasible, search upper half	-
-	new_low	11
12	Binary search iteration	-
-	low	11
-	high	15
-	mid	13
-	check_ok	false
202	mid infeasible, search upper half	-
-	new_low	14
13	Binary search iteration	-
-	low	14
-	high	15
-	mid	14
-	check_ok	false

Bước	Mô tả / Biến	Giá trị
203	mid infeasible, search upper half	-
-	new_low	15
14	Binary search iteration	-
-	low	15
-	high	15
-	mid	15
-	check_ok	true
104	mid feasible, search lower half	-
-	new_high	14
-	best_ans	15
999	Binary search finished	-
-	ans	15
-	check_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	4
-	k	4
-	a	(7, 7, 7, 7)
-	low_initial	7
-	high_initial	28
11	Binary search iteration	-
-	low	7
-	high	28
-	mid	17
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	16
-	best_ans	17
12	Binary search iteration	-
-	low	7
-	high	16
-	mid	11
-	check_ok	true
102	mid feasible, search lower half	-
-	new_high	10
-	best_ans	11
13	Binary search iteration	-
-	low	7
-	high	10
-	mid	8

Bước	Mô tả / Biến	Giá trị
-	check_ok	true
103	mid feasible, search lower half	-
-	new_high	7
-	best_ans	8
14	Binary search iteration	-
-	low	7
-	high	7
-	mid	7
-	check_ok	true
104	mid feasible, search lower half	-
-	new_high	6
-	best_ans	7
999	Binary search finished	-
-	ans	7
-	check_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	5
-	k	2
-	a	(100, 1, 1, 1, 1)
-	low_initial	100
-	high_initial	104
11	Binary search iteration	-
-	low	100
-	high	104
-	mid	102
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	101
-	best_ans	102
12	Binary search iteration	-
-	low	100
-	high	101
-	mid	100
-	check_ok	true
102	mid feasible, search lower half	-
-	new_high	99
-	best_ans	100
999	Binary search finished	-
-	ans	100

Bước	Mô tả / Biến	Giá trị
-	check_count	2

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	6
-	k	3
-	a	(1, 1, 1, 1, 1, 1)
-	low_initial	1
-	high_initial	6
11	Binary search iteration	-
-	low	1
-	high	6
-	mid	3
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	2
-	best_ans	3
12	Binary search iteration	-
-	low	1
-	high	2
-	mid	1
-	check_ok	false
202	mid infeasible, search upper half	-
-	new_low	2
13	Binary search iteration	-
-	low	2
-	high	2
-	mid	2
-	check_ok	true
103	mid feasible, search lower half	-
-	new_high	1
-	best_ans	2
999	Binary search finished	-
-	ans	2
-	check_count	3

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	5
-	k	2
-	a	(3, 2, 4, 1, 5)

Bước	Mô tả / Biến	Giá trị
-	low_initial	5
-	high_initial	15
11	Binary search iteration	-
-	low	5
-	high	15
-	mid	10
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	9
-	best_ans	10
12	Binary search iteration	-
-	low	5
-	high	9
-	mid	7
-	check_ok	false
202	mid infeasible, search upper half	-
-	new_low	8
13	Binary search iteration	-
-	low	8
-	high	9
-	mid	8
-	check_ok	false
203	mid infeasible, search upper half	-
-	new_low	9
14	Binary search iteration	-
-	low	9
-	high	9
-	mid	9
-	check_ok	true
104	mid feasible, search lower half	-
-	new_high	8
-	best_ans	9
999	Binary search finished	-
-	ans	9
-	check_count	4

Bước	Mô tả / Biến	Giá trị
1	Read inputs and compute search bounds	-
-	n	6
-	k	2
-	a	(1, 2, 3, 4, 5, 6)

Bước	Mô tả / Biến	Giá trị
-	low_initial	6
-	high_initial	21
11	Binary search iteration	-
-	low	6
-	high	21
-	mid	13
-	check_ok	true
101	mid feasible, search lower half	-
-	new_high	12
-	best_ans	13
12	Binary search iteration	-
-	low	6
-	high	12
-	mid	9
-	check_ok	false
202	mid infeasible, search upper half	-
-	new_low	10
13	Binary search iteration	-
-	low	10
-	high	12
-	mid	11
-	check_ok	true
103	mid feasible, search lower half	-
-	new_high	10
-	best_ans	11
14	Binary search iteration	-
-	low	10
-	high	10
-	mid	10
-	check_ok	false
204	mid infeasible, search upper half	-
-	new_low	11
999	Binary search finished	-
-	ans	11
-	check_count	4

Lời Giải Chi Tiết Chương 13: Đệ Quy Sơ Cấp & Cây Gọi Đệ Quy

Lời giải & Đáp án: Bài 13.1 - Cây Gọi Đệ Quy: Phân Hoạch & Đếm Nút			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	2	4 3 1	Cây đệ quy nhỏ 3 nút với N=2
2	3	8 5 2	Cây đệ quy 5 nút với N=3
3	4	16 9 3	Cây đệ quy 9 nút với N=4
4	0	1 1 0	Điểm dừng cơ sở N=0 không phân nhánh
5	1	1 1 0	Điểm dừng cơ sở N=1 không phân nhánh
6	5	25 13 4	N=5 với chiều sâu cực đại bằng 4
7	6	39 19 5	N=6 chia đôi thành 3, sinh 19 lời gọi hàm
8	7	54 25 6	N=7 sinh 25 lời gọi hàm và chiều sâu 6
9	8	78 35 7	Tính nhầm và vẽ cây đệ quy N=8 với 35 nút
10	3	8 5 2	Vẽ tay toàn bộ 5 nút và đối chiếu giá trị trả về

Bảng Vết Thực Thi Từng Bước (Chương 13)

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	2
1	Enter recursive call	-
-	n	2
-	depth	0
2	Enter recursive call	-
-	n	1
-	depth	1
1002	Base case reached: n <= 1	-
3	Enter recursive call	-
-	n	1
-	depth	1
1003	Base case reached: n <= 1	-
2003	Return from recursive step	-
-	n	2
-	res	4
999	Recursion completed	-
-	val	4
-	total_calls	3
-	max_depth	1

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	3
1	Enter recursive call	-
-	n	3
-	depth	0
2	Enter recursive call	-
-	n	2
-	depth	1
3	Enter recursive call	-
-	n	1
-	depth	2
1003	Base case reached: $n \leq 1$	-
4	Enter recursive call	-
-	n	1
-	depth	2
1004	Base case reached: $n \leq 1$	-
2004	Return from recursive step	-
-	n	2
-	res	4
5	Enter recursive call	-
-	n	1
-	depth	1
1005	Base case reached: $n \leq 1$	-
2005	Return from recursive step	-
-	n	3
-	res	8
999	Recursion completed	-
-	val	8
-	total_calls	5
-	max_depth	2

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	4
1	Enter recursive call	-
-	n	4
-	depth	0
2	Enter recursive call	-
-	n	3
-	depth	1
3	Enter recursive call	-

Bước	Mô tả / Biến	Giá trị
-	n	2
-	depth	2
4	Enter recursive call	-
-	n	1
-	depth	3
1004	Base case reached: $n \leq 1$	-
5	Enter recursive call	-
-	n	1
-	depth	3
1005	Base case reached: $n \leq 1$	-
2005	Return from recursive step	-
-	n	2
-	res	4
6	Enter recursive call	-
-	n	1
-	depth	2
1006	Base case reached: $n \leq 1$	-
2006	Return from recursive step	-
-	n	3
-	res	8
7	Enter recursive call	-
-	n	2
-	depth	1
8	Enter recursive call	-
-	n	1
-	depth	2
1008	Base case reached: $n \leq 1$	-
9	Enter recursive call	-
-	n	1
-	depth	2
1009	Base case reached: $n \leq 1$	-
2009	Return from recursive step	-
-	n	2
-	res	4
2009	Return from recursive step	-
-	n	4
-	res	16
999	Recursion completed	-
-	val	16
-	total_calls	9
-	max_depth	3

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	0
1	Enter recursive call	-
-	n	0
-	depth	0
1001	Base case reached: $n \leq 1$	-
999	Recursion completed	-
-	val	1
-	total_calls	1
-	max_depth	0

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	1
1	Enter recursive call	-
-	n	1
-	depth	0
1001	Base case reached: $n \leq 1$	-
999	Recursion completed	-
-	val	1
-	total_calls	1
-	max_depth	0

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	5
1	Enter recursive call	-
-	n	5
-	depth	0
2	Enter recursive call	-
-	n	4
-	depth	1
3	Enter recursive call	-
-	n	3
-	depth	2
4	Enter recursive call	-
-	n	2
-	depth	3
5	Enter recursive call	-
-	n	1
-	depth	4

Bước	Mô tả / Biến	Giá trị
1005	Base case reached: $n \leq 1$	-
6	Enter recursive call	-
-	n	1
-	depth	4
1006	Base case reached: $n \leq 1$	-
2006	Return from recursive step	-
-	n	2
-	res	4
7	Enter recursive call	-
-	n	1
-	depth	3
1007	Base case reached: $n \leq 1$	-
2007	Return from recursive step	-
-	n	3
-	res	8
8	Enter recursive call	-
-	n	2
-	depth	2
9	Enter recursive call	-
-	n	1
-	depth	3
1009	Base case reached: $n \leq 1$	-
10	Enter recursive call	-
-	n	1
-	depth	3
1010	Base case reached: $n \leq 1$	-
2010	Return from recursive step	-
-	n	2
-	res	4
2010	Return from recursive step	-
-	n	4
-	res	16
11	Enter recursive call	-
-	n	2
-	depth	1
12	Enter recursive call	-
-	n	1
-	depth	2
1012	Base case reached: $n \leq 1$	-
13	Enter recursive call	-
-	n	1

Bước	Mô tả / Biến	Giá trị
-	depth	2
1013	Base case reached: $n \leq 1$	-
2013	Return from recursive step	-
-	n	2
-	res	4
2013	Return from recursive step	-
-	n	5
-	res	25
999	Recursion completed	-
-	val	25
-	total_calls	13
-	max_depth	4

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	6
1	Enter recursive call	-
-	n	6
-	depth	0
2	Enter recursive call	-
-	n	5
-	depth	1
3	Enter recursive call	-
-	n	4
-	depth	2
4	Enter recursive call	-
-	n	3
-	depth	3
5	Enter recursive call	-
-	n	2
-	depth	4
6	Enter recursive call	-
-	n	1
-	depth	5
1006	Base case reached: $n \leq 1$	-
7	Enter recursive call	-
-	n	1
-	depth	5
1007	Base case reached: $n \leq 1$	-
2007	Return from recursive step	-
-	n	2

Bước	Mô tả / Biến	Giá trị
-	res	4
8	Enter recursive call	-
-	n	1
-	depth	4
1008	Base case reached: $n \leq 1$	-
2008	Return from recursive step	-
-	n	3
-	res	8
9	Enter recursive call	-
-	n	2
-	depth	3
10	Enter recursive call	-
-	n	1
-	depth	4
1010	Base case reached: $n \leq 1$	-
11	Enter recursive call	-
-	n	1
-	depth	4
1011	Base case reached: $n \leq 1$	-
2011	Return from recursive step	-
-	n	2
-	res	4
2011	Return from recursive step	-
-	n	4
-	res	16
12	Enter recursive call	-
-	n	2
-	depth	2
13	Enter recursive call	-
-	n	1
-	depth	3
1013	Base case reached: $n \leq 1$	-
14	Enter recursive call	-
-	n	1
-	depth	3
1014	Base case reached: $n \leq 1$	-
2014	Return from recursive step	-
-	n	2
-	res	4
2014	Return from recursive step	-
-	n	5

Bước	Mô tả / Biến	Giá trị
-	res	25
15	Enter recursive call	-
-	n	3
-	depth	1
16	Enter recursive call	-
-	n	2
-	depth	2
17	Enter recursive call	-
-	n	1
-	depth	3
1017	Base case reached: $n \leq 1$	-
18	Enter recursive call	-
-	n	1
-	depth	3
1018	Base case reached: $n \leq 1$	-
2018	Return from recursive step	-
-	n	2
-	res	4
19	Enter recursive call	-
-	n	1
-	depth	2
1019	Base case reached: $n \leq 1$	-
2019	Return from recursive step	-
-	n	3
-	res	8
2019	Return from recursive step	-
-	n	6
-	res	39
999	Recursion completed	-
-	val	39
-	total_calls	19
-	max_depth	5

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	7
1	Enter recursive call	-
-	n	7
-	depth	0
2	Enter recursive call	-
-	n	6

Bước	Mô tả / Biến	Giá trị
-	depth	1
3	Enter recursive call	-
-	n	5
-	depth	2
4	Enter recursive call	-
-	n	4
-	depth	3
5	Enter recursive call	-
-	n	3
-	depth	4
6	Enter recursive call	-
-	n	2
-	depth	5
7	Enter recursive call	-
-	n	1
-	depth	6
1007	Base case reached: $n \leq 1$	-
8	Enter recursive call	-
-	n	1
-	depth	6
1008	Base case reached: $n \leq 1$	-
2008	Return from recursive step	-
-	n	2
-	res	4
9	Enter recursive call	-
-	n	1
-	depth	5
1009	Base case reached: $n \leq 1$	-
2009	Return from recursive step	-
-	n	3
-	res	8
10	Enter recursive call	-
-	n	2
-	depth	4
11	Enter recursive call	-
-	n	1
-	depth	5
1011	Base case reached: $n \leq 1$	-
12	Enter recursive call	-
-	n	1
-	depth	5

Bước	Mô tả / Biến	Giá trị
1012	Base case reached: $n \leq 1$	-
2012	Return from recursive step	-
-	n	2
-	res	4
2012	Return from recursive step	-
-	n	4
-	res	16
13	Enter recursive call	-
-	n	2
-	depth	3
14	Enter recursive call	-
-	n	1
-	depth	4
1014	Base case reached: $n \leq 1$	-
15	Enter recursive call	-
-	n	1
-	depth	4
1015	Base case reached: $n \leq 1$	-
2015	Return from recursive step	-
-	n	2
-	res	4
2015	Return from recursive step	-
-	n	5
-	res	25
16	Enter recursive call	-
-	n	3
-	depth	2
17	Enter recursive call	-
-	n	2
-	depth	3
18	Enter recursive call	-
-	n	1
-	depth	4
1018	Base case reached: $n \leq 1$	-
19	Enter recursive call	-
-	n	1
-	depth	4
1019	Base case reached: $n \leq 1$	-
2019	Return from recursive step	-
-	n	2
-	res	4

Bước	Mô tả / Biến	Giá trị
20	Enter recursive call	-
-	n	1
-	depth	3
1020	Base case reached: $n \leq 1$	-
2020	Return from recursive step	-
-	n	3
-	res	8
2020	Return from recursive step	-
-	n	6
-	res	39
21	Enter recursive call	-
-	n	3
-	depth	1
22	Enter recursive call	-
-	n	2
-	depth	2
23	Enter recursive call	-
-	n	1
-	depth	3
1023	Base case reached: $n \leq 1$	-
24	Enter recursive call	-
-	n	1
-	depth	3
1024	Base case reached: $n \leq 1$	-
2024	Return from recursive step	-
-	n	2
-	res	4
25	Enter recursive call	-
-	n	1
-	depth	2
1025	Base case reached: $n \leq 1$	-
2025	Return from recursive step	-
-	n	3
-	res	8
2025	Return from recursive step	-
-	n	7
-	res	54
999	Recursion completed	-
-	val	54
-	total_calls	25
-	max_depth	6

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	8
1	Enter recursive call	-
-	n	8
-	depth	0
2	Enter recursive call	-
-	n	7
-	depth	1
3	Enter recursive call	-
-	n	6
-	depth	2
4	Enter recursive call	-
-	n	5
-	depth	3
5	Enter recursive call	-
-	n	4
-	depth	4
6	Enter recursive call	-
-	n	3
-	depth	5
7	Enter recursive call	-
-	n	2
-	depth	6
8	Enter recursive call	-
-	n	1
-	depth	7
1008	Base case reached: $n \leq 1$	-
9	Enter recursive call	-
-	n	1
-	depth	7
1009	Base case reached: $n \leq 1$	-
2009	Return from recursive step	-
-	n	2
-	res	4
10	Enter recursive call	-
-	n	1
-	depth	6
1010	Base case reached: $n \leq 1$	-
2010	Return from recursive step	-
-	n	3
-	res	8

Bước	Mô tả / Biến	Giá trị
11	Enter recursive call	-
-	n	2
-	depth	5
12	Enter recursive call	-
-	n	1
-	depth	6
1012	Base case reached: $n \leq 1$	-
13	Enter recursive call	-
-	n	1
-	depth	6
1013	Base case reached: $n \leq 1$	-
2013	Return from recursive step	-
-	n	2
-	res	4
2013	Return from recursive step	-
-	n	4
-	res	16
14	Enter recursive call	-
-	n	2
-	depth	4
15	Enter recursive call	-
-	n	1
-	depth	5
1015	Base case reached: $n \leq 1$	-
16	Enter recursive call	-
-	n	1
-	depth	5
1016	Base case reached: $n \leq 1$	-
2016	Return from recursive step	-
-	n	2
-	res	4
2016	Return from recursive step	-
-	n	5
-	res	25
17	Enter recursive call	-
-	n	3
-	depth	3
18	Enter recursive call	-
-	n	2
-	depth	4
19	Enter recursive call	-

Bước	Mô tả / Biến	Giá trị
-	n	1
-	depth	5
1019	Base case reached: $n \leq 1$	-
20	Enter recursive call	-
-	n	1
-	depth	5
1020	Base case reached: $n \leq 1$	-
2020	Return from recursive step	-
-	n	2
-	res	4
21	Enter recursive call	-
-	n	1
-	depth	4
1021	Base case reached: $n \leq 1$	-
2021	Return from recursive step	-
-	n	3
-	res	8
2021	Return from recursive step	-
-	n	6
-	res	39
22	Enter recursive call	-
-	n	3
-	depth	2
23	Enter recursive call	-
-	n	2
-	depth	3
24	Enter recursive call	-
-	n	1
-	depth	4
1024	Base case reached: $n \leq 1$	-
25	Enter recursive call	-
-	n	1
-	depth	4
1025	Base case reached: $n \leq 1$	-
2025	Return from recursive step	-
-	n	2
-	res	4
26	Enter recursive call	-
-	n	1
-	depth	3
1026	Base case reached: $n \leq 1$	-

Bước	Mô tả / Biến	Giá trị
2026	Return from recursive step	-
-	n	3
-	res	8
2026	Return from recursive step	-
-	n	7
-	res	54
27	Enter recursive call	-
-	n	4
-	depth	1
28	Enter recursive call	-
-	n	3
-	depth	2
29	Enter recursive call	-
-	n	2
-	depth	3
30	Enter recursive call	-
-	n	1
-	depth	4
1030	Base case reached: $n \leq 1$	-
31	Enter recursive call	-
-	n	1
-	depth	4
1031	Base case reached: $n \leq 1$	-
2031	Return from recursive step	-
-	n	2
-	res	4
32	Enter recursive call	-
-	n	1
-	depth	3
1032	Base case reached: $n \leq 1$	-
2032	Return from recursive step	-
-	n	3
-	res	8
33	Enter recursive call	-
-	n	2
-	depth	2
34	Enter recursive call	-
-	n	1
-	depth	3
1034	Base case reached: $n \leq 1$	-
35	Enter recursive call	-

Bước	Mô tả / Biến	Giá trị
-	n	1
-	depth	3
1035	Base case reached: $n \leq 1$	-
2035	Return from recursive step	-
-	n	2
-	res	4
2035	Return from recursive step	-
-	n	4
-	res	16
2035	Return from recursive step	-
-	n	8
-	res	78
999	Recursion completed	-
-	val	78
-	total_calls	35
-	max_depth	7

Bước	Mô tả / Biến	Giá trị
1	Read input n	-
-	n	3
1	Enter recursive call	-
-	n	3
-	depth	0
2	Enter recursive call	-
-	n	2
-	depth	1
3	Enter recursive call	-
-	n	1
-	depth	2
1003	Base case reached: $n \leq 1$	-
4	Enter recursive call	-
-	n	1
-	depth	2
1004	Base case reached: $n \leq 1$	-
2004	Return from recursive step	-
-	n	2
-	res	4
5	Enter recursive call	-
-	n	1
-	depth	1
1005	Base case reached: $n \leq 1$	-

Bước	Mô tả / Biến	Giá trị
2005	Return from recursive step	-
-	n	3
-	res	8
999	Recursion completed	-
-	val	8
-	total_calls	5
-	max_depth	2

Lời Giải Chi Tiết Chương 14: Thuật Toán Quay Lui & Nhánh Cận

Lời giải & Đáp án: Bài 14.1 - Quay Lui Nhánh Cận: Bài Toán Xếp Hậu Trên Bàn Cờ N x N			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	4	2 2 4 1 3	Bàn cờ 4x4 với đúng 2 nghiệm đối xứng
2	1	1 1	Bàn cờ 1x1 duy nhất 1 ô đặt hậu
3	5	10 1 3 5 2 4	Bàn cờ 5x5 có 10 cấu hình nghiệm
4	2	0 NONE	Bàn cờ 2x2 vô nghiệm
5	3	0 NONE	Bàn cờ 3x3 vô nghiệm
6	6	4 2 4 6 1 3 5	Bàn cờ 6x6 có 4 cấu hình nghiệm
7	7	40 1 3 5 7 2 4 6	Bàn cờ 7x7 có 40 cấu hình nghiệm
8	8	92 1 5 8 6 3 7 2 4	Bàn cờ chuẩn 8x8 có 92 cấu hình nghiệm kinh điển
9	4	2 2 4 1 3	Chạy bàn quay lui từng nước đặt hậu trên bàn cờ 4x4
10	5	10 1 3 5 2 4	Chạy bàn nhánh cận tìm nghiệm đầu tiên của bàn cờ 5x5

Bảng Vết Thực Thi Từng Bước (Chương 14)

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	4
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
521	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2

Bước	Mô tả / Biến	Giá trị
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
1001	Found valid queen placement	-
532	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
1002	Found valid queen placement	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
522	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
511	Backtrack: remove queen from (row, col)	-
503	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	2

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	1
10	Place queen at (row, col)	-
-	row	1
-	col	1
1001	Found valid queen placement	-
500	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	1

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	5
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
1001	Found valid queen placement	-
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
524	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
1002	Found valid queen placement	-
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
521	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	5
1003	Found valid queen placement	-
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
533	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
1004	Found valid queen placement	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
1005	Found valid queen placement	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
1006	Found valid queen placement	-
540	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
1007	Found valid queen placement	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-
-	row	3
-	col	5
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
1008	Found valid queen placement	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
503	Backtrack: remove queen from (row, col)	-
14	Place queen at (row, col)	-
-	row	1
-	col	5
20	Place queen at (row, col)	-
-	row	2
-	col	1

Bước	Mô tả / Biến	Giá trị
33	Place queen at (row, col)	-
-	row	3
-	col	4
523	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
1009	Found valid queen placement	-
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
1010	Found valid queen placement	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
504	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-

Bước	Mô tả / Biến	Giá trị
-	solution_count	10

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	2
10	Place queen at (row, col)	-
-	row	1
-	col	1
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
501	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	0

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	3
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
512	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
510	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	0

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	6
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3

Bước	Mô tả / Biến	Giá trị
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
42	Place queen at (row, col)	-
-	row	4
-	col	3
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
1001	Found valid queen placement	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
515	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
1002	Found valid queen placement	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6

Bước	Mô tả / Biến	Giá trị
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
1003	Found valid queen placement	-
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-
-	row	3
-	col	5
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
503	Backtrack: remove queen from (row, col)	-
14	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	1
-	col	5
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	3
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2

Bước	Mô tả / Biến	Giá trị
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
1004	Found valid queen placement	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
504	Backtrack: remove queen from (row, col)	-
15	Place queen at (row, col)	-
-	row	1
-	col	6
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4

Bước	Mô tả / Biến	Giá trị
533	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
44	Place queen at (row, col)	-
-	row	4
-	col	5
534	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
505	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-

Bước	Mô tả / Biến	Giá trị
-	solution_count	4

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	7
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
1001	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4

Bước	Mô tả / Biến	Giá trị
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5

Bước	Mô tả / Biến	Giá trị
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
42	Place queen at (row, col)	-
-	row	4
-	col	3
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
1002	Found valid queen placement	-
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
1003	Found valid queen placement	-
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
1004	Found valid queen placement	-
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
31	Place queen at (row, col)	-
-	row	3
-	col	2
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
1005	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
76	Place queen at (row, col)	-
-	row	7
-	col	7
1006	Found valid queen placement	-
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
532	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
1007	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
1008	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
1009	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
1010	Found valid queen placement	-
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
26	Place queen at (row, col)	-
-	row	2
-	col	7
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1

Bước	Mô tả / Biến	Giá trị
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
1011	Found valid queen placement	-
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
1012	Found valid queen placement	-
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
1013	Found valid queen placement	-
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	2
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
70	Place queen at (row, col)	-
-	row	7
-	col	1
1014	Found valid queen placement	-
560	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
1015	Found valid queen placement	-
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	5
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
1016	Found valid queen placement	-
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
1017	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
1018	Found valid queen placement	-
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
1019	Found valid queen placement	-

Bước	Mô tả / Biến	Giá trị
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-
-	row	3
-	col	5
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
75	Place queen at (row, col)	-
-	row	7
-	col	6
1020	Found valid queen placement	-
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
1021	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
26	Place queen at (row, col)	-
-	row	2
-	col	7
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	3
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
70	Place queen at (row, col)	-
-	row	7
-	col	1
1022	Found valid queen placement	-
560	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
1023	Found valid queen placement	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
503	Backtrack: remove queen from (row, col)	-
14	Place queen at (row, col)	-
-	row	1
-	col	5
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	6
71	Place queen at (row, col)	-
-	row	7
-	col	2
1024	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
1025	Found valid queen placement	-
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
70	Place queen at (row, col)	-
-	row	7
-	col	1
1026	Found valid queen placement	-
560	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4

Bước	Mô tả / Biến	Giá trị
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
1027	Found valid queen placement	-
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
30	Place queen at (row, col)	-
-	row	3
-	col	1

Bước	Mô tả / Biến	Giá trị
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
55	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
1028	Found valid queen placement	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
1029	Found valid queen placement	-
563	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
504	Backtrack: remove queen from (row, col)	-
15	Place queen at (row, col)	-
-	row	1
-	col	6
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
1030	Found valid queen placement	-
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
1031	Found valid queen placement	-
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
1032	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
1033	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4

Bước	Mô tả / Biến	Giá trị
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
1034	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
44	Place queen at (row, col)	-
-	row	4
-	col	5
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
70	Place queen at (row, col)	-
-	row	7
-	col	1
1035	Found valid queen placement	-
560	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
1036	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
505	Backtrack: remove queen from (row, col)	-
16	Place queen at (row, col)	-
-	row	1
-	col	7
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
1037	Found valid queen placement	-
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
1038	Found valid queen placement	-
563	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
1039	Found valid queen placement	-
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
1040	Found valid queen placement	-
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
506	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	40

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	8

Bước	Mô tả / Biến	Giá trị
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5

Bước	Mô tả / Biến	Giá trị
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6

Bước	Mô tả / Biến	Giá trị
42	Place queen at (row, col)	-
-	row	4
-	col	3
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
36	Place queen at (row, col)	-
-	row	3
-	col	7
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3

Bước	Mô tả / Biến	Giá trị
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
42	Place queen at (row, col)	-
-	row	4
-	col	3
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
45	Place queen at (row, col)	-
-	row	4
-	col	6

Bước	Mô tả / Biến	Giá trị
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7

Bước	Mô tả / Biến	Giá trị
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
83	Place queen at (row, col)	-
-	row	8
-	col	4
1001	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	7
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
84	Place queen at (row, col)	-
-	row	8
-	col	5
1002	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
31	Place queen at (row, col)	-
-	row	3
-	col	2

Bước	Mô tả / Biến	Giá trị
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5

Bước	Mô tả / Biến	Giá trị
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
82	Place queen at (row, col)	-
-	row	8
-	col	3
1003	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	8
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8

Bước	Mô tả / Biến	Giá trị
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1004	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
27	Place queen at (row, col)	-
-	row	2
-	col	8
31	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2

Bước	Mô tả / Biến	Giá trị
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
517	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
54	Place queen at (row, col)	-
-	row	5
-	col	5

Bước	Mô tả / Biến	Giá trị
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
76	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	7
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
76	Place queen at (row, col)	-
-	row	7
-	col	7
84	Place queen at (row, col)	-
-	row	8
-	col	5
1005	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
76	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	7
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	6
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	8
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
83	Place queen at (row, col)	-
-	row	8
-	col	4
1006	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1007	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	3
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6
-	col	8
72	Place queen at (row, col)	-
-	row	7
-	col	3
84	Place queen at (row, col)	-
-	row	8
-	col	5
1008	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4

Bước	Mô tả / Biến	Giá trị
67	Place queen at (row, col)	-
-	row	6
-	col	8
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
84	Place queen at (row, col)	-
-	row	8
-	col	5
1009	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
30	Place queen at (row, col)	-
-	row	3
-	col	1

Bước	Mô tả / Biến	Giá trị
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
57	Place queen at (row, col)	-
-	row	5
-	col	8
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
70	Place queen at (row, col)	-
-	row	7
-	col	1
83	Place queen at (row, col)	-
-	row	8
-	col	4
1010	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1011	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
27	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	2
-	col	8
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
34	Place queen at (row, col)	-
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
76	Place queen at (row, col)	-
-	row	7
-	col	7
83	Place queen at (row, col)	-
-	row	8
-	col	4

Bước	Mô tả / Biến	Giá trị
1012	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
517	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
54	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4

Bước	Mô tả / Biến	Giá trị
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
85	Place queen at (row, col)	-
-	row	8
-	col	6
1013	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
85	Place queen at (row, col)	-
-	row	8
-	col	6
1014	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
76	Place queen at (row, col)	-
-	row	7
-	col	7
80	Place queen at (row, col)	-
-	row	8
-	col	1
1015	Found valid queen placement	-
570	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
77	Place queen at (row, col)	-
-	row	7
-	col	8
85	Place queen at (row, col)	-
-	row	8
-	col	6
1016	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
77	Place queen at (row, col)	-
-	row	7
-	col	8

Bước	Mô tả / Biến	Giá trị
567	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4

Bước	Mô tả / Biến	Giá trị
65	Place queen at (row, col)	-
-	row	6
-	col	6
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
85	Place queen at (row, col)	-
-	row	8
-	col	6
1017	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
25	Place queen at (row, col)	-
-	row	2
-	col	6
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
76	Place queen at (row, col)	-
-	row	7
-	col	7
83	Place queen at (row, col)	-
-	row	8
-	col	4
1018	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
77	Place queen at (row, col)	-
-	row	7
-	col	8
84	Place queen at (row, col)	-
-	row	8
-	col	5
1019	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
77	Place queen at (row, col)	-
-	row	7
-	col	8
83	Place queen at (row, col)	-
-	row	8

Bước	Mô tả / Biến	Giá trị
-	col	4
1020	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
76	Place queen at (row, col)	-
-	row	7
-	col	7
81	Place queen at (row, col)	-
-	row	8
-	col	2
1021	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
76	Place queen at (row, col)	-
-	row	7
-	col	7
80	Place queen at (row, col)	-
-	row	8
-	col	1
1022	Found valid queen placement	-
570	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
81	Place queen at (row, col)	-
-	row	8
-	col	2
1023	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
83	Place queen at (row, col)	-
-	row	8
-	col	4
1024	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
76	Place queen at (row, col)	-
-	row	7
-	col	7
84	Place queen at (row, col)	-
-	row	8
-	col	5
1025	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1

Bước	Mô tả / Biến	Giá trị
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
85	Place queen at (row, col)	-
-	row	8

Bước	Mô tả / Biến	Giá trị
-	col	6
1026	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
70	Place queen at (row, col)	-
-	row	7
-	col	1
84	Place queen at (row, col)	-
-	row	8
-	col	5
1027	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2

Bước	Mô tả / Biến	Giá trị
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
70	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
27	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	2
-	col	8
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6
-	col	6

Bước	Mô tả / Biến	Giá trị
71	Place queen at (row, col)	-
-	row	7
-	col	2
84	Place queen at (row, col)	-
-	row	8
-	col	5
1028	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
517	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1029	Found valid queen placement	-

Bước	Mô tả / Biến	Giá trị
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
81	Place queen at (row, col)	-
-	row	8
-	col	2
1030	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
531	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	3
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	5
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
86	Place queen at (row, col)	-
-	row	8
-	col	7
1031	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
67	Place queen at (row, col)	-
-	row	6
-	col	8
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
67	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	8
70	Place queen at (row, col)	-
-	row	7
-	col	1
84	Place queen at (row, col)	-
-	row	8
-	col	5
1032	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
74	Place queen at (row, col)	-
-	row	7
-	col	5
80	Place queen at (row, col)	-
-	row	8
-	col	1
1033	Found valid queen placement	-
570	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1034	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
557	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
540	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1035	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
75	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
86	Place queen at (row, col)	-
-	row	8
-	col	7
1036	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
67	Place queen at (row, col)	-
-	row	6
-	col	8
72	Place queen at (row, col)	-
-	row	7
-	col	3
86	Place queen at (row, col)	-
-	row	8
-	col	7
1037	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3

Bước	Mô tả / Biến	Giá trị
84	Place queen at (row, col)	-
-	row	8
-	col	5
1038	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
74	Place queen at (row, col)	-
-	row	7
-	col	5
81	Place queen at (row, col)	-
-	row	8
-	col	2
1039	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2

Bước	Mô tả / Biến	Giá trị
-	col	7
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1040	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2

Bước	Mô tả / Biến	Giá trị
64	Place queen at (row, col)	-
-	row	6
-	col	5
70	Place queen at (row, col)	-
-	row	7
-	col	1
85	Place queen at (row, col)	-
-	row	8
-	col	6
1041	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	3
87	Place queen at (row, col)	-
-	row	8
-	col	8
1042	Found valid queen placement	-
577	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6
-	col	6
77	Place queen at (row, col)	-
-	row	7
-	col	8
81	Place queen at (row, col)	-
-	row	8
-	col	2
1043	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	1
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
27	Place queen at (row, col)	-
-	row	2
-	col	8
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
84	Place queen at (row, col)	-
-	row	8

Bước	Mô tả / Biến	Giá trị
-	col	5
1044	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1045	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
65	Place queen at (row, col)	-
-	row	6
-	col	6
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	2
85	Place queen at (row, col)	-
-	row	8
-	col	6
1046	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
517	Backtrack: remove queen from (row, col)	-
503	Backtrack: remove queen from (row, col)	-
14	Place queen at (row, col)	-
-	row	1
-	col	5
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	7
82	Place queen at (row, col)	-
-	row	8
-	col	3
1047	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1048	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
83	Place queen at (row, col)	-
-	row	8
-	col	4
1049	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
70	Place queen at (row, col)	-
-	row	7
-	col	1
86	Place queen at (row, col)	-
-	row	8
-	col	7
1050	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6

Bước	Mô tả / Biến	Giá trị
80	Place queen at (row, col)	-
-	row	8
-	col	1
1051	Found valid queen placement	-
570	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
77	Place queen at (row, col)	-
-	row	7
-	col	8
82	Place queen at (row, col)	-
-	row	8
-	col	3
1052	Found valid queen placement	-

Bước	Mô tả / Biến	Giá trị
572	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1053	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
70	Place queen at (row, col)	-
-	row	7
-	col	1

Bước	Mô tả / Biến	Giá trị
560	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3

Bước	Mô tả / Biến	Giá trị
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
86	Place queen at (row, col)	-
-	row	8
-	col	7
1054	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	6
83	Place queen at (row, col)	-
-	row	8
-	col	4
1055	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
75	Place queen at (row, col)	-
-	row	7
-	col	6
81	Place queen at (row, col)	-
-	row	8
-	col	2
1056	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
26	Place queen at (row, col)	-
-	row	2
-	col	7
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
73	Place queen at (row, col)	-
-	row	7
-	col	4
81	Place queen at (row, col)	-
-	row	8
-	col	2
1057	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
67	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	8
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1058	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
67	Place queen at (row, col)	-
-	row	6
-	col	8
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	8
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1059	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
87	Place queen at (row, col)	-
-	row	8
-	col	8
1060	Found valid queen placement	-
577	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
77	Place queen at (row, col)	-
-	row	7
-	col	8
83	Place queen at (row, col)	-
-	row	8
-	col	4

Bước	Mô tả / Biến	Giá trị
1061	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
75	Place queen at (row, col)	-
-	row	7
-	col	6
81	Place queen at (row, col)	-
-	row	8
-	col	2

Bước	Mô tả / Biến	Giá trị
1062	Found valid queen placement	-
571	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
516	Backtrack: remove queen from (row, col)	-
27	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	2
-	col	8
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	7
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2

Bước	Mô tả / Biến	Giá trị
551	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
71	Place queen at (row, col)	-
-	row	7
-	col	2
86	Place queen at (row, col)	-
-	row	8
-	col	7
1063	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7

Bước	Mô tả / Biến	Giá trị
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
82	Place queen at (row, col)	-
-	row	8
-	col	3
1064	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3

Bước	Mô tả / Biến	Giá trị
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
56	Place queen at (row, col)	-
-	row	5
-	col	7
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
517	Backtrack: remove queen from (row, col)	-
504	Backtrack: remove queen from (row, col)	-
15	Place queen at (row, col)	-
-	row	1
-	col	6
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
76	Place queen at (row, col)	-
-	row	7
-	col	7
83	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	8
-	col	4
1065	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3

Bước	Mô tả / Biến	Giá trị
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	4
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	7
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
77	Place queen at (row, col)	-
-	row	7
-	col	8
83	Place queen at (row, col)	-
-	row	8
-	col	4
1066	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	8
74	Place queen at (row, col)	-
-	row	7
-	col	5
82	Place queen at (row, col)	-
-	row	8
-	col	3
1067	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
67	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5

Bước	Mô tả / Biến	Giá trị
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
71	Place queen at (row, col)	-
-	row	7
-	col	2
83	Place queen at (row, col)	-
-	row	8
-	col	4
1068	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
84	Place queen at (row, col)	-
-	row	8
-	col	5
1069	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
86	Place queen at (row, col)	-
-	row	8
-	col	7
1070	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
520	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
87	Place queen at (row, col)	-
-	row	8
-	col	8
1071	Found valid queen placement	-
577	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
86	Place queen at (row, col)	-
-	row	8
-	col	7
1072	Found valid queen placement	-
576	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
76	Place queen at (row, col)	-
-	row	7
-	col	7
566	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	8
70	Place queen at (row, col)	-
-	row	7
-	col	1
84	Place queen at (row, col)	-
-	row	8
-	col	5
1073	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
70	Place queen at (row, col)	-
-	row	7
-	col	1
83	Place queen at (row, col)	-
-	row	8
-	col	4
1074	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
71	Place queen at (row, col)	-
-	row	7
-	col	2
84	Place queen at (row, col)	-
-	row	8
-	col	5
1075	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	1
44	Place queen at (row, col)	-
-	row	4
-	col	5
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
82	Place queen at (row, col)	-
-	row	8
-	col	3
1076	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
77	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	8
567	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
70	Place queen at (row, col)	-
-	row	7
-	col	1
560	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
70	Place queen at (row, col)	-
-	row	7
-	col	1
82	Place queen at (row, col)	-
-	row	8
-	col	3
1077	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
560	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
87	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	8
-	col	8
1078	Found valid queen placement	-
577	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
82	Place queen at (row, col)	-
-	row	8
-	col	3
1079	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
27	Place queen at (row, col)	-
-	row	2
-	col	8
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	5
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
82	Place queen at (row, col)	-
-	row	8
-	col	3
1080	Found valid queen placement	-
572	Backtrack: remove queen from (row, col)	-
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7

Bước	Mô tả / Biến	Giá trị
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
46	Place queen at (row, col)	-
-	row	4
-	col	7
53	Place queen at (row, col)	-
-	row	5
-	col	4
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
66	Place queen at (row, col)	-
-	row	6
-	col	7

Bước	Mô tả / Biến	Giá trị
556	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
517	Backtrack: remove queen from (row, col)	-
505	Backtrack: remove queen from (row, col)	-
16	Place queen at (row, col)	-
-	row	1
-	col	7
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
84	Place queen at (row, col)	-
-	row	8
-	col	5
1081	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6

Bước	Mô tả / Biến	Giá trị
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1082	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
540	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3

Bước	Mô tả / Biến	Giá trị
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
77	Place queen at (row, col)	-
-	row	7
-	col	8
84	Place queen at (row, col)	-
-	row	8
-	col	5
1083	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
567	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8

Bước	Mô tả / Biến	Giá trị
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
50	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
64	Place queen at (row, col)	-
-	row	6
-	col	5
71	Place queen at (row, col)	-
-	row	7
-	col	2
83	Place queen at (row, col)	-
-	row	8
-	col	4
1084	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
73	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
73	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	4
563	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
75	Place queen at (row, col)	-
-	row	7
-	col	6
83	Place queen at (row, col)	-
-	row	8
-	col	4
1085	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
565	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
60	Place queen at (row, col)	-
-	row	6
-	col	1
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
57	Place queen at (row, col)	-
-	row	5
-	col	8
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
62	Place queen at (row, col)	-
-	row	6
-	col	3
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
57	Place queen at (row, col)	-
-	row	5
-	col	8
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1086	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
547	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
60	Place queen at (row, col)	-
-	row	6
-	col	1
72	Place queen at (row, col)	-
-	row	7
-	col	3
84	Place queen at (row, col)	-
-	row	8
-	col	5
1087	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
550	Backtrack: remove queen from (row, col)	-
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
35	Place queen at (row, col)	-
-	row	3
-	col	6
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
525	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
47	Place queen at (row, col)	-
-	row	4
-	col	8
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
77	Place queen at (row, col)	-
-	row	7
-	col	8
567	Backtrack: remove queen from (row, col)	-
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
50	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
67	Place queen at (row, col)	-
-	row	6
-	col	8
557	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6

Bước	Mô tả / Biến	Giá trị
63	Place queen at (row, col)	-
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
67	Place queen at (row, col)	-
-	row	6
-	col	8
71	Place queen at (row, col)	-
-	row	7
-	col	2
83	Place queen at (row, col)	-
-	row	8
-	col	4
1088	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
561	Backtrack: remove queen from (row, col)	-
557	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
47	Place queen at (row, col)	-
-	row	4
-	col	8
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6

Bước	Mô tả / Biến	Giá trị
63	Place queen at (row, col)	-
-	row	6
-	col	4
71	Place queen at (row, col)	-
-	row	7
-	col	2
561	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
537	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
37	Place queen at (row, col)	-
-	row	3
-	col	8
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7

Bước	Mô tả / Biến	Giá trị
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
62	Place queen at (row, col)	-
-	row	6
-	col	3
552	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
53	Place queen at (row, col)	-
-	row	5
-	col	4
543	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
527	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
506	Backtrack: remove queen from (row, col)	-
17	Place queen at (row, col)	-
-	row	1
-	col	8
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
66	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	7
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
52	Place queen at (row, col)	-
-	row	5

Bước	Mô tả / Biến	Giá trị
-	col	3
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2

Bước	Mô tả / Biến	Giá trị
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	2
-	col	2
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
85	Place queen at (row, col)	-
-	row	8
-	col	6
1089	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
542	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
40	Place queen at (row, col)	-
-	row	4
-	col	1
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
85	Place queen at (row, col)	-
-	row	8
-	col	6
1090	Found valid queen placement	-
575	Backtrack: remove queen from (row, col)	-
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
76	Place queen at (row, col)	-
-	row	7
-	col	7
83	Place queen at (row, col)	-
-	row	8

Bước	Mô tả / Biến	Giá trị
-	col	4
1091	Found valid queen placement	-
573	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
41	Place queen at (row, col)	-
-	row	4
-	col	2

Bước	Mô tả / Biến	Giá trị
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
51	Place queen at (row, col)	-
-	row	5
-	col	2
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
66	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	7
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
76	Place queen at (row, col)	-
-	row	7
-	col	7
84	Place queen at (row, col)	-
-	row	8
-	col	5
1092	Found valid queen placement	-
574	Backtrack: remove queen from (row, col)	-
566	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
65	Place queen at (row, col)	-
-	row	6
-	col	6
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
555	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
61	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
46	Place queen at (row, col)	-
-	row	4
-	col	7
52	Place queen at (row, col)	-
-	row	5
-	col	3
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
64	Place queen at (row, col)	-
-	row	6
-	col	5
554	Backtrack: remove queen from (row, col)	-
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
42	Place queen at (row, col)	-
-	row	4
-	col	3
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
74	Place queen at (row, col)	-
-	row	7
-	col	5
564	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
545	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
535	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
50	Place queen at (row, col)	-
-	row	5
-	col	1
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
52	Place queen at (row, col)	-
-	row	5
-	col	3
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
535	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3

Bước	Mô tả / Biến	Giá trị
40	Place queen at (row, col)	-
-	row	4
-	col	1
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
61	Place queen at (row, col)	-
-	row	6
-	col	2
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
551	Backtrack: remove queen from (row, col)	-
63	Place queen at (row, col)	-
-	row	6
-	col	4
75	Place queen at (row, col)	-
-	row	7
-	col	6
565	Backtrack: remove queen from (row, col)	-
553	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
45	Place queen at (row, col)	-
-	row	4
-	col	6
535	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
36	Place queen at (row, col)	-
-	row	3
-	col	7
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
65	Place queen at (row, col)	-
-	row	6
-	col	6
555	Backtrack: remove queen from (row, col)	-
542	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6
61	Place queen at (row, col)	-
-	row	6
-	col	2
551	Backtrack: remove queen from (row, col)	-
545	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-
-	row	4
-	col	2
55	Place queen at (row, col)	-
-	row	5
-	col	6
545	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
55	Place queen at (row, col)	-
-	row	5
-	col	6

Bước	Mô tả / Biến	Giá trị
545	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
526	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
25	Place queen at (row, col)	-
-	row	2
-	col	6
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
66	Place queen at (row, col)	-
-	row	6
-	col	7
73	Place queen at (row, col)	-
-	row	7
-	col	4
563	Backtrack: remove queen from (row, col)	-
556	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
546	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
541	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
536	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
31	Place queen at (row, col)	-
-	row	3
-	col	2
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
60	Place queen at (row, col)	-
-	row	6
-	col	1
550	Backtrack: remove queen from (row, col)	-
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
56	Place queen at (row, col)	-
-	row	5
-	col	7
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
64	Place queen at (row, col)	-
-	row	6

Bước	Mô tả / Biến	Giá trị
-	col	5
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
51	Place queen at (row, col)	-
-	row	5
-	col	2
63	Place queen at (row, col)	-
-	row	6
-	col	4
553	Backtrack: remove queen from (row, col)	-
541	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
41	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
56	Place queen at (row, col)	-
-	row	5
-	col	7
64	Place queen at (row, col)	-
-	row	6
-	col	5
72	Place queen at (row, col)	-
-	row	7
-	col	3
562	Backtrack: remove queen from (row, col)	-
554	Backtrack: remove queen from (row, col)	-
546	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
46	Place queen at (row, col)	-
-	row	4
-	col	7
50	Place queen at (row, col)	-
-	row	5
-	col	1
540	Backtrack: remove queen from (row, col)	-
54	Place queen at (row, col)	-
-	row	5
-	col	5
544	Backtrack: remove queen from (row, col)	-
536	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
515	Backtrack: remove queen from (row, col)	-
507	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	92

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	4
10	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
521	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
23	Place queen at (row, col)	-
-	row	2
-	col	4
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
1001	Found valid queen placement	-
532	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4

Bước	Mô tả / Biến	Giá trị
41	Place queen at (row, col)	-
-	row	4
-	col	2
1002	Found valid queen placement	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
522	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
511	Backtrack: remove queen from (row, col)	-
503	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	2

Bước	Mô tả / Biến	Giá trị
1	Read board size n	-
-	n	5
10	Place queen at (row, col)	-
-	row	1
-	col	1
22	Place queen at (row, col)	-
-	row	2
-	col	3
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4

Bước	Mô tả / Biến	Giá trị
-	col	2
53	Place queen at (row, col)	-
-	row	5
-	col	4
1001	Found valid queen placement	-
543	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
23	Place queen at (row, col)	-
-	row	2
-	col	4
31	Place queen at (row, col)	-
-	row	3
-	col	2
44	Place queen at (row, col)	-
-	row	4
-	col	5
52	Place queen at (row, col)	-
-	row	5
-	col	3
1002	Found valid queen placement	-
542	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
521	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
500	Backtrack: remove queen from (row, col)	-
11	Place queen at (row, col)	-
-	row	1
-	col	2
23	Place queen at (row, col)	-
-	row	2
-	col	4

Bước	Mô tả / Biến	Giá trị
30	Place queen at (row, col)	-
-	row	3
-	col	1
42	Place queen at (row, col)	-
-	row	4
-	col	3
54	Place queen at (row, col)	-
-	row	5
-	col	5
1003	Found valid queen placement	-
544	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
513	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
533	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
32	Place queen at (row, col)	-
-	row	3
-	col	3
40	Place queen at (row, col)	-
-	row	4
-	col	1
53	Place queen at (row, col)	-
-	row	5
-	col	4
1004	Found valid queen placement	-
543	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
501	Backtrack: remove queen from (row, col)	-
12	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	1
-	col	3
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
41	Place queen at (row, col)	-
-	row	4
-	col	2
54	Place queen at (row, col)	-
-	row	5
-	col	5
1005	Found valid queen placement	-
544	Backtrack: remove queen from (row, col)	-
531	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
24	Place queen at (row, col)	-
-	row	2
-	col	5
31	Place queen at (row, col)	-
-	row	3
-	col	2
43	Place queen at (row, col)	-
-	row	4
-	col	4
50	Place queen at (row, col)	-
-	row	5
-	col	1
1006	Found valid queen placement	-
540	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
521	Backtrack: remove queen from (row, col)	-
514	Backtrack: remove queen from (row, col)	-
502	Backtrack: remove queen from (row, col)	-
13	Place queen at (row, col)	-
-	row	1
-	col	4
20	Place queen at (row, col)	-

Bước	Mô tả / Biến	Giá trị
-	row	2
-	col	1
32	Place queen at (row, col)	-
-	row	3
-	col	3
44	Place queen at (row, col)	-
-	row	4
-	col	5
51	Place queen at (row, col)	-
-	row	5
-	col	2
1007	Found valid queen placement	-
541	Backtrack: remove queen from (row, col)	-
534	Backtrack: remove queen from (row, col)	-
522	Backtrack: remove queen from (row, col)	-
34	Place queen at (row, col)	-
-	row	3
-	col	5
41	Place queen at (row, col)	-
-	row	4
-	col	2
531	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
34	Place queen at (row, col)	-
-	row	3
-	col	5
42	Place queen at (row, col)	-
-	row	4
-	col	3
50	Place queen at (row, col)	-
-	row	5
-	col	1
1008	Found valid queen placement	-
540	Backtrack: remove queen from (row, col)	-
532	Backtrack: remove queen from (row, col)	-
524	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-

Bước	Mô tả / Biến	Giá trị
503	Backtrack: remove queen from (row, col)	-
14	Place queen at (row, col)	-
-	row	1
-	col	5
20	Place queen at (row, col)	-
-	row	2
-	col	1
33	Place queen at (row, col)	-
-	row	3
-	col	4
523	Backtrack: remove queen from (row, col)	-
510	Backtrack: remove queen from (row, col)	-
21	Place queen at (row, col)	-
-	row	2
-	col	2
33	Place queen at (row, col)	-
-	row	3
-	col	4
40	Place queen at (row, col)	-
-	row	4
-	col	1
52	Place queen at (row, col)	-
-	row	5
-	col	3
1009	Found valid queen placement	-
542	Backtrack: remove queen from (row, col)	-
530	Backtrack: remove queen from (row, col)	-
523	Backtrack: remove queen from (row, col)	-
511	Backtrack: remove queen from (row, col)	-
22	Place queen at (row, col)	-
-	row	2
-	col	3
30	Place queen at (row, col)	-
-	row	3
-	col	1
43	Place queen at (row, col)	-
-	row	4
-	col	4
51	Place queen at (row, col)	-
-	row	5
-	col	2

Bước	Mô tả / Biến	Giá trị
1010	Found valid queen placement	-
541	Backtrack: remove queen from (row, col)	-
533	Backtrack: remove queen from (row, col)	-
520	Backtrack: remove queen from (row, col)	-
512	Backtrack: remove queen from (row, col)	-
504	Backtrack: remove queen from (row, col)	-
999	Finished backtracking search	-
-	solution_count	10

Lời Giải Chi Tiết Chương 15: Thuật Toán Tham Lam & Phương Pháp Chứng Minh Trên Giấy

Lời giải & Đáp án: Bài 15.1 - Thuật Toán Tham Lam: Lựa Chọn Hoạt Động Tối Đa			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	4 1 3 2 5 3 9 6 8	2 4	Bốn cuộc họp mẫu kinh điển
2	3 1 2 2 3 3 4	3 3	Các cuộc họp nối tiếp nhau liên tục
3	4 1 10 2 3 4 5 6 7	3 3	Một cuộc họp dài bao phủ 3 cuộc họp ngắn
4	1 5 10	1 5	Một cuộc họp duy nhất
5	3 1 4 5 8 9 12	3 9	Các cuộc họp hoàn toàn cách xa nhau
6	3 2 6 2 6 2 6	1 4	Tất cả cuộc họp có cùng khung giờ
7	5 1 10 2 9 3 8 4 7 5 6	1 1	Các khoảng thời gian lồng nhau, chọn khoảng trong cùng
8	3 0 5 5 10 10 15	3 15	Biên điểm mút chạm nhau $S_j = E_i$ vẫn tương thích
9	8 1 4 3 5 0 6 5 7 3 8 5 9 6 10 8 11	3 8	Tập 8 khoảng thời gian tính nhầm bằng Exchange Argument

Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
10	5 2 5 1 3 4 7 6 9 8 10	3 7	Sắp xếp 5 khoảng thời gian không theo thứ tự ban đầu

Bảng Vết Thực Thi Từng Bước (Chương 15)

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	4
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	1
-	end	3
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	3
-	total_duration	2
11	Consider interval (s, e)	-
-	id	2
-	start	2
-	end	5
-	last_end	3
201	Conflict: Discarded this meeting	-
12	Consider interval (s, e)	-
-	id	4
-	start	6
-	end	8
-	last_end	3
102	Compatible: Selected this meeting	-
-	current_count	2
-	new_last_end	8
-	total_duration	4
13	Consider interval (s, e)	-
-	id	3
-	start	3
-	end	9
-	last_end	8
203	Conflict: Discarded this meeting	-

Bước	Mô tả / Biến	Giá trị
999	Greedy activity selection complete	-
-	max_meetings	2
-	total_duration	4

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	3
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	1
-	end	2
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	2
-	total_duration	1
11	Consider interval (s, e)	-
-	id	2
-	start	2
-	end	3
-	last_end	2
101	Compatible: Selected this meeting	-
-	current_count	2
-	new_last_end	3
-	total_duration	2
12	Consider interval (s, e)	-
-	id	3
-	start	3
-	end	4
-	last_end	3
102	Compatible: Selected this meeting	-
-	current_count	3
-	new_last_end	4
-	total_duration	3
999	Greedy activity selection complete	-
-	max_meetings	3
-	total_duration	3

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-

Bước	Mô tả / Biến	Giá trị
-	n	4
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	2
-	start	2
-	end	3
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	3
-	total_duration	1
11	Consider interval (s, e)	-
-	id	3
-	start	4
-	end	5
-	last_end	3
101	Compatible: Selected this meeting	-
-	current_count	2
-	new_last_end	5
-	total_duration	2
12	Consider interval (s, e)	-
-	id	4
-	start	6
-	end	7
-	last_end	5
102	Compatible: Selected this meeting	-
-	current_count	3
-	new_last_end	7
-	total_duration	3
13	Consider interval (s, e)	-
-	id	1
-	start	1
-	end	10
-	last_end	7
203	Conflict: Discarded this meeting	-
999	Greedy activity selection complete	-
-	max_meetings	3
-	total_duration	3

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-

Bước	Mô tả / Biến	Giá trị
-	n	1
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	5
-	end	10
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	10
-	total_duration	5
999	Greedy activity selection complete	-
-	max_meetings	1
-	total_duration	5

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	3
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	1
-	end	4
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	4
-	total_duration	3
11	Consider interval (s, e)	-
-	id	2
-	start	5
-	end	8
-	last_end	4
101	Compatible: Selected this meeting	-
-	current_count	2
-	new_last_end	8
-	total_duration	6
12	Consider interval (s, e)	-
-	id	3
-	start	9
-	end	12

Bước	Mô tả / Biến	Giá trị
-	last_end	8
102	Compatible: Selected this meeting	-
-	current_count	3
-	new_last_end	12
-	total_duration	9
999	Greedy activity selection complete	-
-	max_meetings	3
-	total_duration	9

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	3
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	2
-	end	6
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	6
-	total_duration	4
11	Consider interval (s, e)	-
-	id	2
-	start	2
-	end	6
-	last_end	6
201	Conflict: Discarded this meeting	-
12	Consider interval (s, e)	-
-	id	3
-	start	2
-	end	6
-	last_end	6
202	Conflict: Discarded this meeting	-
999	Greedy activity selection complete	-
-	max_meetings	1
-	total_duration	4

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	5

Bước	Mô tả / Biến	Giá trị
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	5
-	start	5
-	end	6
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	6
-	total_duration	1
11	Consider interval (s, e)	-
-	id	4
-	start	4
-	end	7
-	last_end	6
201	Conflict: Discarded this meeting	-
12	Consider interval (s, e)	-
-	id	3
-	start	3
-	end	8
-	last_end	6
202	Conflict: Discarded this meeting	-
13	Consider interval (s, e)	-
-	id	2
-	start	2
-	end	9
-	last_end	6
203	Conflict: Discarded this meeting	-
14	Consider interval (s, e)	-
-	id	1
-	start	1
-	end	10
-	last_end	6
204	Conflict: Discarded this meeting	-
999	Greedy activity selection complete	-
-	max_meetings	1
-	total_duration	1

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	3

Bước	Mô tả / Biến	Giá trị
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	0
-	end	5
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	5
-	total_duration	5
11	Consider interval (s, e)	-
-	id	2
-	start	5
-	end	10
-	last_end	5
101	Compatible: Selected this meeting	-
-	current_count	2
-	new_last_end	10
-	total_duration	10
12	Consider interval (s, e)	-
-	id	3
-	start	10
-	end	15
-	last_end	10
102	Compatible: Selected this meeting	-
-	current_count	3
-	new_last_end	15
-	total_duration	15
999	Greedy activity selection complete	-
-	max_meetings	3
-	total_duration	15

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	8
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	1
-	start	1
-	end	4
-	last_end	-1

Bước	Mô tả / Biến	Giá trị
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	4
-	total_duration	3
11	Consider interval (s, e)	-
-	id	2
-	start	3
-	end	5
-	last_end	4
201	Conflict: Discarded this meeting	-
12	Consider interval (s, e)	-
-	id	3
-	start	0
-	end	6
-	last_end	4
202	Conflict: Discarded this meeting	-
13	Consider interval (s, e)	-
-	id	4
-	start	5
-	end	7
-	last_end	4
103	Compatible: Selected this meeting	-
-	current_count	2
-	new_last_end	7
-	total_duration	5
14	Consider interval (s, e)	-
-	id	5
-	start	3
-	end	8
-	last_end	7
204	Conflict: Discarded this meeting	-
15	Consider interval (s, e)	-
-	id	6
-	start	5
-	end	9
-	last_end	7
205	Conflict: Discarded this meeting	-
16	Consider interval (s, e)	-
-	id	7
-	start	6
-	end	10

Bước	Mô tả / Biến	Giá trị
-	last_end	7
206	Conflict: Discarded this meeting	-
17	Consider interval (s, e)	-
-	id	8
-	start	8
-	end	11
-	last_end	7
107	Compatible: Selected this meeting	-
-	current_count	3
-	new_last_end	11
-	total_duration	8
999	Greedy activity selection complete	-
-	max_meetings	3
-	total_duration	8

Bước	Mô tả / Biến	Giá trị
1	Read input intervals	-
-	n	5
2	Sorted intervals by earliest finish time	-
10	Consider interval (s, e)	-
-	id	2
-	start	1
-	end	3
-	last_end	-1
100	Compatible: Selected this meeting	-
-	current_count	1
-	new_last_end	3
-	total_duration	2
11	Consider interval (s, e)	-
-	id	1
-	start	2
-	end	5
-	last_end	3
201	Conflict: Discarded this meeting	-
12	Consider interval (s, e)	-
-	id	3
-	start	4
-	end	7
-	last_end	3
102	Compatible: Selected this meeting	-
-	current_count	2

Bước	Mô tả / Biến	Giá trị
-	new_last_end	7
-	total_duration	5
13	Consider interval (s, e)	-
-	id	4
-	start	6
-	end	9
-	last_end	7
203	Conflict: Discarded this meeting	-
14	Consider interval (s, e)	-
-	id	5
-	start	8
-	end	10
-	last_end	7
104	Compatible: Selected this meeting	-
-	current_count	3
-	new_last_end	10
-	total_duration	7
999	Greedy activity selection complete	-
-	max_meetings	3
-	total_duration	7

Lời Giải Chi Tiết Chương 16: Ngăn Xếp & Hàng Đợi STL

Lời giải & Đáp án: Bài 16.1 - Ngăn Xếp Đơn Điều: Phần Tử Lớn Hơn Đầu Tiên Bên Phải			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	4 4 5 2 25	5 25 25 -1	Mẫu kinh điển của bài toán Next Greater Element
2	4 13 7 6 12	-1 12 12 -1	Phần tử đầu tiên lớn nhất không có phần tử nào bên phải lớn hơn
3	4 1 2 3 4	2 3 4 -1	Dãy tăng dần, mỗi phần tử có NGE là phần tử liền kề
4	1 5	-1	Mảng 1 phần tử duy nhất luôn trả về -1
5	5 5 4 3 2 1	-1 -1 -1 -1 -1	Dãy giảm dần ngặt, không phần tử nào có NGE
6	4 7 7 7 7	-1 -1 -1 -1	Các phần tử bằng nhau hoàn toàn
7	5 10 3 10 3 10	-1 10 -1 10 -1	Giá trị cực đại lặp lại xen kẽ số nhỏ
8	5 1 5 2 8 3	5 8 8 -1 -1	Phần tử lớn ở giữa che khuất các phần tử bên phải
9	6 3 1 4 1 5 9	4 4 5 5 9 -1	Dãy chữ số số Pi tính nhẩm trên giấy qua 6 bước
10	7 6 2 5 4 5 1 6	-1 5 6 5 6 6 -1	Dãy ziczac với nhiều lần pop liên tiếp trên ngăn xếp

Bảng Vết Thực Thi Từng Bước (Chương 16)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	4
-	a	(4, 5, 2, 25)
10	Examine element a[i] from right	-
-	i	3
-	a[i]	25
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	25
11	Examine element a[i] from right	-
-	i	2
-	a[i]	2
201	Found next greater element on stack top	-
-	nge[i]	25
-	pushed_to_stack	2

Bước	Mô tả / Biến	Giá trị
12	Examine element a[i] from right	-
-	i	1
-	a[i]	5
102	Pop smaller or equal element from stack	-
-	popped_top	2
202	Found next greater element on stack top	-
-	nge[i]	25
-	pushed_to_stack	5
13	Examine element a[i] from right	-
-	i	0
-	a[i]	4
203	Found next greater element on stack top	-
-	nge[i]	5
-	pushed_to_stack	4
999	Finished Monotonic Stack scan	-
-	nge	(5, 25, 25, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	4
-	a	(13, 7, 6, 12)
10	Examine element a[i] from right	-
-	i	3
-	a[i]	12
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	12
11	Examine element a[i] from right	-
-	i	2
-	a[i]	6
201	Found next greater element on stack top	-
-	nge[i]	12
-	pushed_to_stack	6
12	Examine element a[i] from right	-
-	i	1
-	a[i]	7
102	Pop smaller or equal element from stack	-
-	popped_top	6
202	Found next greater element on stack top	-
-	nge[i]	12
-	pushed_to_stack	7

Bước	Mô tả / Biến	Giá trị
13	Examine element a[i] from right	-
-	i	0
-	a[i]	13
103	Pop smaller or equal element from stack	-
-	popped_top	7
103	Pop smaller or equal element from stack	-
-	popped_top	12
303	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	13
999	Finished Monotonic Stack scan	-
-	nge	(-1, 12, 12, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	4
-	a	(1, 2, 3, 4)
10	Examine element a[i] from right	-
-	i	3
-	a[i]	4
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	4
11	Examine element a[i] from right	-
-	i	2
-	a[i]	3
201	Found next greater element on stack top	-
-	nge[i]	4
-	pushed_to_stack	3
12	Examine element a[i] from right	-
-	i	1
-	a[i]	2
202	Found next greater element on stack top	-
-	nge[i]	3
-	pushed_to_stack	2
13	Examine element a[i] from right	-
-	i	0
-	a[i]	1
203	Found next greater element on stack top	-
-	nge[i]	2
-	pushed_to_stack	1

Bước	Mô tả / Biến	Giá trị
999	Finished Monotonic Stack scan	-
-	nge	(2, 3, 4, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	1
-	a	(5,)
10	Examine element a[i] from right	-
-	i	0
-	a[i]	5
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	5
999	Finished Monotonic Stack scan	-
-	nge	(-1,)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	5
-	a	(5, 4, 3, 2, 1)
10	Examine element a[i] from right	-
-	i	4
-	a[i]	1
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	1
11	Examine element a[i] from right	-
-	i	3
-	a[i]	2
101	Pop smaller or equal element from stack	-
-	popped_top	1
301	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	2
12	Examine element a[i] from right	-
-	i	2
-	a[i]	3
102	Pop smaller or equal element from stack	-
-	popped_top	2
302	No greater element on right	-
-	nge[i]	-1

Bước	Mô tả / Biến	Giá trị
-	pushed_to_stack	3
13	Examine element a[i] from right	-
-	i	1
-	a[i]	4
103	Pop smaller or equal element from stack	-
-	popped_top	3
303	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	4
14	Examine element a[i] from right	-
-	i	0
-	a[i]	5
104	Pop smaller or equal element from stack	-
-	popped_top	4
304	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	5
999	Finished Monotonic Stack scan	-
-	nge	(-1, -1, -1, -1, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	4
-	a	(7, 7, 7, 7)
10	Examine element a[i] from right	-
-	i	3
-	a[i]	7
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	7
11	Examine element a[i] from right	-
-	i	2
-	a[i]	7
101	Pop smaller or equal element from stack	-
-	popped_top	7
301	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	7
12	Examine element a[i] from right	-
-	i	1
-	a[i]	7

Bước	Mô tả / Biến	Giá trị
102	Pop smaller or equal element from stack	-
-	popped_top	7
302	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	7
13	Examine element a[i] from right	-
-	i	0
-	a[i]	7
103	Pop smaller or equal element from stack	-
-	popped_top	7
303	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	7
999	Finished Monotonic Stack scan	-
-	nge	(-1, -1, -1, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	5
-	a	(10, 3, 10, 3, 10)
10	Examine element a[i] from right	-
-	i	4
-	a[i]	10
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	10
11	Examine element a[i] from right	-
-	i	3
-	a[i]	3
201	Found next greater element on stack top	-
-	nge[i]	10
-	pushed_to_stack	3
12	Examine element a[i] from right	-
-	i	2
-	a[i]	10
102	Pop smaller or equal element from stack	-
-	popped_top	3
102	Pop smaller or equal element from stack	-
-	popped_top	10
302	No greater element on right	-
-	nge[i]	-1

Bước	Mô tả / Biến	Giá trị
-	pushed_to_stack	10
13	Examine element a[i] from right	-
-	i	1
-	a[i]	3
203	Found next greater element on stack top	-
-	nge[i]	10
-	pushed_to_stack	3
14	Examine element a[i] from right	-
-	i	0
-	a[i]	10
104	Pop smaller or equal element from stack	-
-	popped_top	3
104	Pop smaller or equal element from stack	-
-	popped_top	10
304	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	10
999	Finished Monotonic Stack scan	-
-	nge	(-1, 10, -1, 10, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	5
-	a	(1, 5, 2, 8, 3)
10	Examine element a[i] from right	-
-	i	4
-	a[i]	3
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	3
11	Examine element a[i] from right	-
-	i	3
-	a[i]	8
101	Pop smaller or equal element from stack	-
-	popped_top	3
301	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	8
12	Examine element a[i] from right	-
-	i	2
-	a[i]	2

Bước	Mô tả / Biến	Giá trị
202	Found next greater element on stack top	-
-	nge[i]	8
-	pushed_to_stack	2
13	Examine element a[i] from right	-
-	i	1
-	a[i]	5
103	Pop smaller or equal element from stack	-
-	popped_top	2
203	Found next greater element on stack top	-
-	nge[i]	8
-	pushed_to_stack	5
14	Examine element a[i] from right	-
-	i	0
-	a[i]	1
204	Found next greater element on stack top	-
-	nge[i]	5
-	pushed_to_stack	1
999	Finished Monotonic Stack scan	-
-	nge	(5, 8, 8, -1, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	6
-	a	(3, 1, 4, 1, 5, 9)
10	Examine element a[i] from right	-
-	i	5
-	a[i]	9
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	9
11	Examine element a[i] from right	-
-	i	4
-	a[i]	5
201	Found next greater element on stack top	-
-	nge[i]	9
-	pushed_to_stack	5
12	Examine element a[i] from right	-
-	i	3
-	a[i]	1
202	Found next greater element on stack top	-
-	nge[i]	5

Bước	Mô tả / Biến	Giá trị
-	pushed_to_stack	1
13	Examine element a[i] from right	-
-	i	2
-	a[i]	4
103	Pop smaller or equal element from stack	-
-	popped_top	1
203	Found next greater element on stack top	-
-	nge[i]	5
-	pushed_to_stack	4
14	Examine element a[i] from right	-
-	i	1
-	a[i]	1
204	Found next greater element on stack top	-
-	nge[i]	4
-	pushed_to_stack	1
15	Examine element a[i] from right	-
-	i	0
-	a[i]	3
105	Pop smaller or equal element from stack	-
-	popped_top	1
205	Found next greater element on stack top	-
-	nge[i]	4
-	pushed_to_stack	3
999	Finished Monotonic Stack scan	-
-	nge	(4, 4, 5, 5, 9, -1)

Bước	Mô tả / Biến	Giá trị
1	Read inputs	-
-	n	7
-	a	(6, 2, 5, 4, 5, 1, 6)
10	Examine element a[i] from right	-
-	i	6
-	a[i]	6
300	No greater element on right	-
-	nge[i]	-1
-	pushed_to_stack	6
11	Examine element a[i] from right	-
-	i	5
-	a[i]	1
201	Found next greater element on stack top	-
-	nge[i]	6

Bước	Mô tả / Biến	Giá trị
-	pushed_to_stack	1
12	Examine element a[i] from right	-
-	i	4
-	a[i]	5
102	Pop smaller or equal element from stack	-
-	popped_top	1
202	Found next greater element on stack top	-
-	nge[i]	6
-	pushed_to_stack	5
13	Examine element a[i] from right	-
-	i	3
-	a[i]	4
203	Found next greater element on stack top	-
-	nge[i]	5
-	pushed_to_stack	4
14	Examine element a[i] from right	-
-	i	2
-	a[i]	5
104	Pop smaller or equal element from stack	-
-	popped_top	4
104	Pop smaller or equal element from stack	-
-	popped_top	5
204	Found next greater element on stack top	-
-	nge[i]	6
-	pushed_to_stack	5
15	Examine element a[i] from right	-
-	i	1
-	a[i]	2
205	Found next greater element on stack top	-
-	nge[i]	5
-	pushed_to_stack	2
16	Examine element a[i] from right	-
-	i	0
-	a[i]	6
106	Pop smaller or equal element from stack	-
-	popped_top	2
106	Pop smaller or equal element from stack	-
-	popped_top	5
106	Pop smaller or equal element from stack	-
-	popped_top	6
306	No greater element on right	-

Bước	Mô tả / Biến	Giá trị
-	nge[i]	-1
-	pushed_to_stack	6
999	Finished Monotonic Stack scan	-
-	nge	(-1, 5, 6, 5, 6, 6, -1)

Lời Giải Chi Tiết Chương 17: Tập Hợp & Ánh Xạ STL

Lời giải & Đáp án: Bài 17.1 - Tập Hợp & Ánh Xạ: Thống Kê Tần Suất & Truy Vấn Ngưỡng Cận			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	6 1 10 1 20 1 10 2 15 2 25 1 30	20 -1 3 10	Truy vấn chèn và tìm lower_bound cơ bản
2	4 1 5 1 15 2 5 2 6	5 15 2 5	Tìm đúng phần tử bằng và phần tử lớn hơn kế tiếp
3	4 1 100 2 50 2 100 2 150	100 100 -1 1 100	Một phần tử duy nhất được truy vấn nhiều ngưỡng
4	2 2 10 2 20	-1 -1 0 0	Tập hợp rỗng khi chỉ nhận truy vấn tìm kiếm
5	3 1 7 1 7 1 7	1 7	Chèn cùng một giá trị nhiều lần chỉ có 1 phần tử duy nhất
6	4 1 3 1 8 2 3 2 8	3 8 2 3	Các truy vấn khớp đúng các giá trị trong tập hợp
7	3 1 20 1 10 2 15	20 2 10	Hòa tần suất chọn phần tử có giá trị nhỏ hơn
8	4 1 -10 1 -5 2 -8 2 0	-5 -1 2 -10	Tập hợp chứa các số nguyên âm

Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
9	7 1 4 1 2 1 6 2 3 1 2 2 5 2 7	4 6 -1 3 2	Theo dõi trạng thái Cây Đồ-Đen qua 7 thao tác
10	7 1 1 1 5 1 3 1 5 2 2 2 4 1 3	3 5 3 3	Cân bằng tần suất giữa 3 và 5, chọn số 3 nhỏ hơn

Bảng Vết Thực Thi Từng Bước (Chương 17)

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	6
10	Process query	-
-	type	1
-	x	10
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	20
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	1
-	x	10
102	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	2
13	Process query	-
-	type	2
-	x	15
203	Queried lower_bound(x)	-
-	lb_res	20

Bước	Mô tả / Biến	Giá trị
14	Process query	-
-	type	2
-	x	25
204	Queried lower_bound(x)	-
-	lb_res	-1
15	Process query	-
-	type	1
-	x	30
105	Inserted into set and increased frequency	-
-	set_size	3
-	freq_x	1
999	Finished query stream	-
-	distinct_count	3
-	max_freq_val	10
-	max_count	2

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	4
10	Process query	-
-	type	1
-	x	5
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	15
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	2
-	x	5
202	Queried lower_bound(x)	-
-	lb_res	5
13	Process query	-
-	type	2
-	x	6
203	Queried lower_bound(x)	-
-	lb_res	15

Bước	Mô tả / Biến	Giá trị
999	Finished query stream	-
-	distinct_count	2
-	max_freq_val	5
-	max_count	1

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	4
10	Process query	-
-	type	1
-	x	100
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	2
-	x	50
201	Queried lower_bound(x)	-
-	lb_res	100
12	Process query	-
-	type	2
-	x	100
202	Queried lower_bound(x)	-
-	lb_res	100
13	Process query	-
-	type	2
-	x	150
203	Queried lower_bound(x)	-
-	lb_res	-1
999	Finished query stream	-
-	distinct_count	1
-	max_freq_val	100
-	max_count	1

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	2
10	Process query	-
-	type	2
-	x	10
200	Queried lower_bound(x)	-

Bước	Mô tả / Biến	Giá trị
-	lb_res	-1
11	Process query	-
-	type	2
-	x	20
201	Queried lower_bound(x)	-
-	lb_res	-1
999	Set is empty	-

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	3
10	Process query	-
-	type	1
-	x	7
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	7
101	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	2
12	Process query	-
-	type	1
-	x	7
102	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	3
999	Finished query stream	-
-	distinct_count	1
-	max_freq_val	7
-	max_count	3

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	4
10	Process query	-
-	type	1
-	x	3
100	Inserted into set and increased frequency	-

Bước	Mô tả / Biến	Giá trị
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	8
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	2
-	x	3
202	Queried lower_bound(x)	-
-	lb_res	3
13	Process query	-
-	type	2
-	x	8
203	Queried lower_bound(x)	-
-	lb_res	8
999	Finished query stream	-
-	distinct_count	2
-	max_freq_val	3
-	max_count	1

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	3
10	Process query	-
-	type	1
-	x	20
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	10
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	2
-	x	15

Bước	Mô tả / Biến	Giá trị
202	Queried lower_bound(x)	-
-	lb_res	20
999	Finished query stream	-
-	distinct_count	2
-	max_freq_val	10
-	max_count	1

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	4
10	Process query	-
-	type	1
-	x	-10
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	-5
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	2
-	x	-8
202	Queried lower_bound(x)	-
-	lb_res	-5
13	Process query	-
-	type	2
-	x	0
203	Queried lower_bound(x)	-
-	lb_res	-1
999	Finished query stream	-
-	distinct_count	2
-	max_freq_val	-10
-	max_count	1

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	7
10	Process query	-

Bước	Mô tả / Biến	Giá trị
-	type	1
-	x	4
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	2
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	1
-	x	6
102	Inserted into set and increased frequency	-
-	set_size	3
-	freq_x	1
13	Process query	-
-	type	2
-	x	3
203	Queried lower_bound(x)	-
-	lb_res	4
14	Process query	-
-	type	1
-	x	2
104	Inserted into set and increased frequency	-
-	set_size	3
-	freq_x	2
15	Process query	-
-	type	2
-	x	5
205	Queried lower_bound(x)	-
-	lb_res	6
16	Process query	-
-	type	2
-	x	7
206	Queried lower_bound(x)	-
-	lb_res	-1
999	Finished query stream	-
-	distinct_count	3
-	max_freq_val	2

Bước	Mô tả / Biến	Giá trị
-	max_count	2

Bước	Mô tả / Biến	Giá trị
1	Initialize set and map	-
-	n	7
10	Process query	-
-	type	1
-	x	1
100	Inserted into set and increased frequency	-
-	set_size	1
-	freq_x	1
11	Process query	-
-	type	1
-	x	5
101	Inserted into set and increased frequency	-
-	set_size	2
-	freq_x	1
12	Process query	-
-	type	1
-	x	3
102	Inserted into set and increased frequency	-
-	set_size	3
-	freq_x	1
13	Process query	-
-	type	1
-	x	5
103	Inserted into set and increased frequency	-
-	set_size	3
-	freq_x	2
14	Process query	-
-	type	2
-	x	2
204	Queried lower_bound(x)	-
-	lb_res	3
15	Process query	-
-	type	2
-	x	4
205	Queried lower_bound(x)	-
-	lb_res	5
16	Process query	-
-	type	1

Bước	Mô tả / Biến	Giá trị
-	x	3
106	Inserted into set and increased frequency	-
-	set_size	3
-	freq_x	2
999	Finished query stream	-
-	distinct_count	3
-	max_freq_val	3
-	max_count	2

Lời Giải Chi Tiết Bài Tập Tích Hợp: Phân Đoạn & Ghép Cặp Tài Nguyên Tối Ưu

Lời giải & Đáp án: Bài Tập Tích Hợp 2 - Phân Đoạn & Ghép Cặp Tài Nguyên Tối Ưu			
Test	Đầu vào (Input)	Đáp án chính thức	Ghi chú / Phương pháp
1	5 7 2 3 1 4 1 5	4 1 8 8	Bài toán tích hợp mẫu với 5 gói tài nguyên
2	4 10 2 2 4 6 8	4 2 4 12	Dãy số chẵn chia làm 2 cụm máy
3	5 6 3 1 2 3 2 1	3 1 10 3	Dãy đối xứng có 3 phần tử duy nhất
4	1 10 1 5	1 5 0 5	Một phần tử duy nhất, không tạo được cặp
5	4 8 2 4 4 4 4	1 4 6 8	Mọi phần tử đều bằng nhau và ghép đôi vừa vặn
6	4 20 4 10 20 30 40	4 10 0 40	K bằng N, tải trọng bằng phần tử lớn nhất
7	4 3 2 5 6 7 8	4 5 0 15	Ngưỡng S quá nhỏ không tạo được cặp nào
8	3 15 1 2 5 3	3 2 3 10	K=1 máy nhận toàn bộ tải trọng bằng tổng mảng
9	6 10 3 1 5 2 8 3 4	6 1 12 8	Dãy 6 phần tử phân biệt tính nhầm cả 3 công đoạn
10	6 8 2 2 3 2 4 1 3	4 2 15 8	Dãy có tần suất trùng lặp và tính 15 cặp

Bảng Vết Thực Thi Từng Bước (Bài Tập Tích Hợp)

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	5
-	s	7
-	k	2
-	a	(3, 1, 4, 1, 5)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	4
-	max_freq_val	1
3	Phase 2: Sorted array for Two Pointers	-
-	b	(1, 1, 3, 4, 5)
100	Two Pointers: valid pairs found	-
-	left	0
-	right	4
-	added_pairs	4

Bước	Mô tả / Biến	Giá trị
101	Two Pointers: valid pairs found	-
-	left	1
-	right	4
-	added_pairs	3
204	Two Pointers: sum exceeds s, shrink right	-
102	Two Pointers: valid pairs found	-
-	left	2
-	right	3
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	8
301	Phase 3 BS: check(mid)	-
-	mid	9
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	6
-	ok	false
303	Phase 3 BS: check(mid)	-
-	mid	7
-	ok	false
304	Phase 3 BS: check(mid)	-
-	mid	8
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	8

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	4
-	s	10
-	k	2
-	a	(2, 4, 6, 8)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	4
-	max_freq_val	2
3	Phase 2: Sorted array for Two Pointers	-
-	b	(2, 4, 6, 8)
100	Two Pointers: valid pairs found	-
-	left	0
-	right	3

Bước	Mô tả / Biến	Giá trị
-	added_pairs	3
203	Two Pointers: sum exceeds s, shrink right	-
101	Two Pointers: valid pairs found	-
-	left	1
-	right	2
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	4
301	Phase 3 BS: check(mid)	-
-	mid	14
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	10
-	ok	false
303	Phase 3 BS: check(mid)	-
-	mid	12
-	ok	true
304	Phase 3 BS: check(mid)	-
-	mid	11
-	ok	false
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	12

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	5
-	s	6
-	k	3
-	a	(1, 2, 3, 2, 1)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	3
-	max_freq_val	1
3	Phase 2: Sorted array for Two Pointers	-
-	b	(1, 1, 2, 2, 3)
100	Two Pointers: valid pairs found	-
-	left	0
-	right	4
-	added_pairs	4
101	Two Pointers: valid pairs found	-
-	left	1

Bước	Mô tả / Biến	Giá trị
-	right	4
-	added_pairs	3
102	Two Pointers: valid pairs found	-
-	left	2
-	right	4
-	added_pairs	2
103	Two Pointers: valid pairs found	-
-	left	3
-	right	4
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	10
301	Phase 3 BS: check(mid)	-
-	mid	6
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	4
-	ok	true
303	Phase 3 BS: check(mid)	-
-	mid	3
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	3

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	1
-	s	10
-	k	1
-	a	(5,)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	1
-	max_freq_val	5
3	Phase 2: Sorted array for Two Pointers	-
-	b	(5,)
4	Phase 2 completed: Total valid pairs	-
-	pair_count	0
301	Phase 3 BS: check(mid)	-
-	mid	5
-	ok	true

Bước	Mô tả / Biến	Giá trị
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	5

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	4
-	s	8
-	k	2
-	a	(4, 4, 4, 4)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	1
-	max_freq_val	4
3	Phase 2: Sorted array for Two Pointers	-
-	b	(4, 4, 4, 4)
100	Two Pointers: valid pairs found	-
-	left	0
-	right	3
-	added_pairs	3
101	Two Pointers: valid pairs found	-
-	left	1
-	right	3
-	added_pairs	2
102	Two Pointers: valid pairs found	-
-	left	2
-	right	3
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	6
301	Phase 3 BS: check(mid)	-
-	mid	10
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	6
-	ok	false
303	Phase 3 BS: check(mid)	-
-	mid	8
-	ok	true
304	Phase 3 BS: check(mid)	-
-	mid	7
-	ok	false

Bước	Mô tả / Biến	Giá trị
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	8

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	4
-	s	20
-	k	4
-	a	(10, 20, 30, 40)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	4
-	max_freq_val	10
3	Phase 2: Sorted array for Two Pointers	-
-	b	(10, 20, 30, 40)
203	Two Pointers: sum exceeds s, shrink right	-
202	Two Pointers: sum exceeds s, shrink right	-
201	Two Pointers: sum exceeds s, shrink right	-
4	Phase 2 completed: Total valid pairs	-
-	pair_count	0
301	Phase 3 BS: check(mid)	-
-	mid	70
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	54
-	ok	true
303	Phase 3 BS: check(mid)	-
-	mid	46
-	ok	true
304	Phase 3 BS: check(mid)	-
-	mid	42
-	ok	true
305	Phase 3 BS: check(mid)	-
-	mid	40
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	40

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	4
-	s	3

Bước	Mô tả / Biến	Giá trị
-	k	2
-	a	(5, 6, 7, 8)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	4
-	max_freq_val	5
3	Phase 2: Sorted array for Two Pointers	-
-	b	(5, 6, 7, 8)
203	Two Pointers: sum exceeds s, shrink right	-
202	Two Pointers: sum exceeds s, shrink right	-
201	Two Pointers: sum exceeds s, shrink right	-
4	Phase 2 completed: Total valid pairs	-
-	pair_count	0
301	Phase 3 BS: check(mid)	-
-	mid	17
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	12
-	ok	false
303	Phase 3 BS: check(mid)	-
-	mid	14
-	ok	false
304	Phase 3 BS: check(mid)	-
-	mid	15
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	15

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	3
-	s	15
-	k	1
-	a	(2, 5, 3)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	3
-	max_freq_val	2
3	Phase 2: Sorted array for Two Pointers	-
-	b	(2, 3, 5)
100	Two Pointers: valid pairs found	-
-	left	0

Bước	Mô tả / Biến	Giá trị
-	right	2
-	added_pairs	2
101	Two Pointers: valid pairs found	-
-	left	1
-	right	2
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	3
301	Phase 3 BS: check(mid)	-
-	mid	7
-	ok	false
302	Phase 3 BS: check(mid)	-
-	mid	9
-	ok	false
303	Phase 3 BS: check(mid)	-
-	mid	10
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	10

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	6
-	s	10
-	k	3
-	a	(1, 5, 2, 8, 3, 4)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	6
-	max_freq_val	1
3	Phase 2: Sorted array for Two Pointers	-
-	b	(1, 2, 3, 4, 5, 8)
100	Two Pointers: valid pairs found	-
-	left	0
-	right	5
-	added_pairs	5
101	Two Pointers: valid pairs found	-
-	left	1
-	right	5
-	added_pairs	4
205	Two Pointers: sum exceeds s, shrink right	-

Bước	Mô tả / Biến	Giá trị
102	Two Pointers: valid pairs found	-
-	left	2
-	right	4
-	added_pairs	2
103	Two Pointers: valid pairs found	-
-	left	3
-	right	4
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	12
301	Phase 3 BS: check(mid)	-
-	mid	15
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	11
-	ok	true
303	Phase 3 BS: check(mid)	-
-	mid	9
-	ok	true
304	Phase 3 BS: check(mid)	-
-	mid	8
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	8

Bước	Mô tả / Biến	Giá trị
1	Read inputs and populated STL Set and Map	-
-	n	6
-	s	8
-	k	2
-	a	(2, 3, 2, 4, 1, 3)
2	Phase 1: Unique count and max frequency value	-
-	unique_count	4
-	max_freq_val	2
3	Phase 2: Sorted array for Two Pointers	-
-	b	(1, 2, 2, 3, 3, 4)
100	Two Pointers: valid pairs found	-
-	left	0
-	right	5
-	added_pairs	5

Bước	Mô tả / Biến	Giá trị
101	Two Pointers: valid pairs found	-
-	left	1
-	right	5
-	added_pairs	4
102	Two Pointers: valid pairs found	-
-	left	2
-	right	5
-	added_pairs	3
103	Two Pointers: valid pairs found	-
-	left	3
-	right	5
-	added_pairs	2
104	Two Pointers: valid pairs found	-
-	left	4
-	right	5
-	added_pairs	1
4	Phase 2 completed: Total valid pairs	-
-	pair_count	15
301	Phase 3 BS: check(mid)	-
-	mid	9
-	ok	true
302	Phase 3 BS: check(mid)	-
-	mid	6
-	ok	false
303	Phase 3 BS: check(mid)	-
-	mid	7
-	ok	false
304	Phase 3 BS: check(mid)	-
-	mid	8
-	ok	true
5	Phase 3 completed: Min max workload found	-
-	min_max_workload	8